Statistica

Esami e domande aperte

Enunciati e soluzioni complete

Inferenza statistica - Favaro

Raccolta personale

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Chapter 1

17 January 2011

1.1 Question 1

Enunciato. Introduce the likelihood function and describe the maximum likelihood method for constructing estimators. Provide and solve at least one problem in which the maximum likelihood method cannot be applied.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

An example without an attained MLE is an exponential rate model with a single observed value $x = 0$: $L(\theta) = \theta$ grows without bound for $\theta > 0$. This zero observation has probability zero under

the continuous law, but as an optimization problem no maximizing parameter exists. A non-regular moving-support example also has no attained maximum with a literal open support: for $U(0,\theta)$ and observed maximum $M > 0$, $L = \theta^{-n}\mathbf{1}_{\{\theta>M\}}$, whose supremum is approached at M from above. Closing the support yields the conventional endpoint estimate but is an explicit choice of density version.

1.2 Question 2

Enunciato. The response time of a computer to a terminal's input is modeled as an exponential random variable with unknown parameter θ. In an experiment n response times $X_1, \dots, X_n$ are measured. Prove that $\hat{T}_n = n^{-1}\sum X_i$ is an unbiased estimator of $1/\theta$. Is $1/\hat{T}_n$ unbiased for θ? Derive an unbiased estimator S_n of θ, compute its MSE, and determine whether it is consistent. Hint: $n\hat{T}_n$ has density $\theta^n x^{n-1}e^{-\theta x}/(n-1)!$.

Soluzione. The density in the hint identifies the rate convention. Thus $EX_i = 1/\theta$ and $\operatorname{Var} X_i = 1/\theta^2$, giving $E\hat{T}_n = 1/\theta$. Put $T = \sum X_i \sim \text{Gamma}(n,\theta)$. For $n > k$, direct integration gives $ET^{-k} = \theta^k\Gamma(n-k)/\Gamma(n)$. Hence for $n > 1$, $E(1/\hat{T}_n) = E(n/T) = n\theta/(n-1)$, an upward bias; for $n = 1$ the expectation diverges. Set $S_n = (n-1)/T$ for $n > 1$. It is exactly unbiased. For $n > 2$,

$$ES_n^2 = \frac{(n-1)\theta^2}{n-2}, \quad \operatorname{MSE}(S_n) = \operatorname{Var} S_n = \frac{\theta^2}{n-2}.$$

For $n = 2$ its variance is infinite. As $n \to \infty$ the MSE tends to zero, proving consistency; the strong LLN also gives $(n-1)/T = (1-1/n)/\bar{X} \to \theta$ almost surely.

Chapter 2

31 January 2011

2.1 Question 1

Enunciato. The Neyman–Pearson lemma in the theory of statistical hypothesis testing. Provide and solve one example of application of the lemma or its generalizations under a Gaussian assumption for the random sample.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0,1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x)\exp\{\eta(\theta)T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \mathrm{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \leq \alpha \leq P_{\theta_0}(T \geq k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t}\theta^t(1-\theta)^{n-t} + \gamma\binom{n}{k}\theta^k(1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

For iid $N(\mu, \sigma^2)$ with known σ, testing $\mu = \mu_0$ against $\mu = \mu_1 > \mu_0$ gives log ratio $(\mu_1 - \mu_0)\sum X_i/\sigma^2 - n(\mu_1^2 - \mu_0^2)/(2\sigma^2)$, increasing in $\bar{X}$. Reject if $\bar{X} > \mu_0 + \sigma z_{1-\alpha}/\sqrt{n}$. Under μ_0 this has size α; its power at μ is $1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu - \mu_0)/\sigma)$. The same region is MP for each larger mean, hence UMP for the one-sided composite alternative.

2.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample with $f_X(x;\theta) = \theta x^{\theta-1}\mathbf{1}_{(0,1)}(x)$. Find the maximum likelihood estimator of θ, a sufficient statistic, and the moment-based estimator. Check unbiasedness and consistency of both estimators.

Soluzione. Put $T = -\sum \log X_i$. The likelihood is $\theta^n e^{-(\theta-1)T}$, so T is sufficient. Differentiation gives $\hat{\theta}_{ML} = n/T$. The variables $-\log X_i$ are iid $\text{Exp}(\theta)$ and $T \sim \text{Gamma}(n, \theta)$. For $n > 1$, $E\hat{\theta}_{ML} = n\theta/(n-1)$, so it is biased; for $n = 1$ the expectation is infinite. The first raw moment is $\theta/(\theta+1)$, giving $\hat{\theta}_{MM} = \bar{X}/(1-\bar{X})$. This function is strictly convex on $(0,1)$, hence its expectation is strictly above θ whenever finite (and is not unbiased if infinite). Both estimators converge strongly to θ by the strong LLN and continuity. Their Gaussian limits are respectively variance θ^2 and $\theta(\theta+1)^2/(\theta+2)$ after multiplication by $\sqrt{n}$: the first follows by Delta applied to $1/\bar{Z}$ and the second by Delta applied to $\bar{X}/(1-\bar{X})$.

Chapter 3

14 February 2011

3.1 Question 1

Enunciato. Sufficiency in statistical inference based on the likelihood function: definitions, main properties, theorems and examples.

Soluzione. A statistic T is sufficient for θ if the conditional law of the complete sample given T is independent of θ. In a dominated model the Fisher–Neyman factorization theorem says this is equivalent to

$$f_\theta(x) = g_\theta(T(x))h(x).$$

In the discrete case, for $T = t$ the conditional probability is $g_\theta(t)h(x)/\sum_{T(y)=t} g_\theta(t)h(y)$, so the parameter cancels. Conversely a parameter-free conditional law times the marginal law of T gives the factorization. In the dominated continuous case use conditional distributions and disintegration; the same cancellation holds almost everywhere.

For Bernoulli or Poisson sampling, $\sum X_i$ is sufficient. Gaussian sampling with known variance has sufficient $\sum X_i$ for the mean; with both mean and variance unknown $(\sum X_i, \sum X_i^2)$ is sufficient. For an exponential family $f_\theta = h \exp\{\eta(\theta)A - \psi(\theta)\}$, the sample sum of A is sufficient.

A statistic is minimal sufficient if it is a measurable function of every sufficient statistic, up to common null sets. In a dominated positive model, the likelihood-ratio criterion states that T is minimal sufficient when $T(x) = T(y)$ iff $f_\theta(x)/f_\theta(y)$ is independent of θ. Any sufficient factorization forces that parameter-free ratio within its fibers; therefore its partition refines that of T. Conversely the ratio criterion provides a factorization by choosing representatives (with standard measurability conditions). For Gaussian mean with known variance the ratio depends on θ exactly through $\sum x_i - \sum y_i$, so the sample sum is minimal sufficient. This theorem concerns minimal sufficiency and is distinct from the completeness-based UMVU theorem also associated with Lehmann and Scheffé.

Rao–Blackwell: for a square-integrable estimator U, let $V = E[U \mid T]$ with sufficient T. The conditional expectation can be chosen independently of the unknown parameter, so V is an estimator; $EV = EU$. Variance decomposition gives $\operatorname{Var} U = \operatorname{Var} V + E \operatorname{Var}(U \mid T) \geq \operatorname{Var} V$. For any target $g(\theta)$, conditioning also gives smaller squared-error risk. In Bernoulli sampling start with $U = X_1$, an unbiased estimator of θ. Given $T = \sum X_i = t$, exchangeability gives $V = t/n = \bar{X}$. Its variance is $\theta(1-\theta)/n$, strictly smaller than that of X_1 for $n > 1$ and $0 < \theta < 1$; it attains the Cramér–Rao bound. Completeness of the binomial statistic further makes it the unique UMVU estimator.

3.2 Question 2

Enunciato. The weekly number of arrests for failure in a department has a Poisson distribution with parameter λ, and counts in subsequent weeks are independent. Let $X_1, \ldots, X_{10}$ be the counts

in ten weeks and $\bar{X} = 10^{-1}\sum X_i$. Determine t so that $\{\bar{X} > t\}$ is a level-0.05 rejection region for $H_0 : \lambda = 2$ versus $H_1 : \lambda = 3$. What is its power? What threshold and power apply with twenty weeks? What is the rejection region under the Neyman–Pearson likelihood approach?

Soluzione. For m weeks, $T = \sum X_i \sim \text{Pois}(m\lambda)$. The log likelihood ratio is $T\log(3/2) - m$, increasing in T, so NP gives an upper-tail test. Choose the smallest integer k_m with $P_{2m}(T \geq k_m) \leq 0.05$; then take $t_m = (k_m - 1)/m$. Its exact power is $P_{3m}(T \geq k_m)$, computed by $1 - e^{-3m}\sum_{j=0}^{k_m-1}(3m)^j/j!$. For ten weeks $k_{10} = 29$, $t_{10} = 2.8$, size approximately 0.03433 and power 0.59692. For twenty weeks $k_{20} = 52$, $t_{20} = 2.55$, size approximately 0.03874 and power 0.86509. Because the count is discrete a deterministic rule may have size strictly below 0.05. Exact size is obtained by rejecting for $T \geq k_m$ and randomizing at $T = k_m - 1$ with probability $[0.05 - P_{2m}(T \geq k_m)]/P_{2m}(T = k_m - 1)$. Its power adds that randomization probability times $P_{3m}(T = k_m - 1)$.

Chapter 4

19 December 2011

4.1 Question 1

Enunciato. Introduce the likelihood function and describe the maximum likelihood method applied to the theory of point estimation. Provide and solve at least one problem of point estimation in which the usual conditions of regularity for the likelihood function are not satisfied.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these

conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0,\theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat{\theta} = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \mathrm{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \le u) = u$. Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \mathrm{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b, a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

4.2 Question 2

Enunciato. In a biological population only the 40% of the individuals survive after the first week of life. The population census after one week detects k individuals. We want to estimate the number N of hatched eggs. Which is the maximum likelihood estimator $\hat{N}$ of N? Compute the value of the estimator for $k = 400$.

Soluzione. The sampling model is $K \sim \mathrm{Bin}(N,p)$ with known $p = 0.4$, integer $N \ge k$. For $k > 0$,

$$L(N) = \binom{N}{k}p^k(1-p)^{N-k}, \quad \frac{L(N+1)}{L(N)} = (1-p)\frac{N+1}{N+1-k}.$$

This ratio is at least one exactly when $N + 1 \le k/p$. If k/p is not an integer, the unique maximum is $\lfloor k/p \rfloor$; if $k/p = m$ is an integer, both $m-1$ and m maximize (provided $m - 1 \ge k$). For $k = 400$, $m = 1000$, so the MLE is not unique: 999 and 1000 have equal largest likelihood. Reporting $1000 = k/p$ is an allowed choice. If $k = 0$, likelihood $(1-p)^N$ decreases and the maximum is the smallest allowed N (zero if permitted).

Chapter 5

17 December 2012

5.1 Question 1

Enunciato. Describe finite and asymptotic efficiency for an estimator T_n of θ. State and prove a necessary and sufficient condition for existence of an efficient and unbiased estimator.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T-\theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

5.2 Question 2

Enunciato. Let $(X_1, \ldots, X_n)$ have density $f_X(x; \theta) = e^{-(x-\theta)}\mathbf{1}_{(\theta, +\infty)}(x)$. Find the MLE of θ. Check unbiasedness and absolute efficiency, at finite sample size and asymptotically. Identify its parametric distribution if possible. Provide an optimal size-α rejection region for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.

Soluzione. Put $M = X_{(1)}$. The likelihood is $L(\theta) = e^{-\sum x_i + n\theta}\mathbf{1}_{\{\theta < M\}}$. It increases throughout its admissible parameter set. With the literal strict support it has no attained maximum; its supremum

is approached as $\theta \uparrow M$. The conventional endpoint MLE, using an equivalent density version with $x \geq \theta$, is $\hat{\theta} = M$. Writing $X_i = \theta + E_i$, where E_i are iid Exp(1), gives $M - \theta \sim \text{Exp}(n)$. Hence

$$E\hat{\theta} = \theta + 1/n, \quad \text{Var}\,\hat{\theta} = 1/n^2, \quad \text{MSE}\,\hat{\theta} = 2/n^2.$$

It is biased at finite n, asymptotically unbiased and mean-square consistent. The estimator $M - 1/n$ is unbiased. For $H_0 : \theta = \theta_0$ versus $\theta_1 > \theta_0$, the likelihood ratio is zero if $M < \theta_1$ and constant $e^{n(\theta_1 - \theta_0)}$ on $M \geq \theta_1$. Thus rejecting in a largest-M region is NP optimal. Choose

$$c = \theta_0 - \frac{\log \alpha}{n}, \qquad C = \{M \geq c\}.$$

Under the null $P(C) = e^{-n(c-\theta_0)} = \alpha$. Under $\theta > \theta_0$ its power is $e^{-n(c-\theta)}$ if $\theta < c$, and one if $\theta \geq c$. If $c < \theta_1$, the region contains the entire alternative support and has power one, so it is still most powerful. The same region is optimal for every positive shift, hence is UMP for the one-sided composite alternative.

The distribution is a shifted exponential: $\hat{\theta} - \theta \sim \text{Exp}(n)$. The regular CR efficiency definition is inapplicable to its moving support, and the MLE is not an unbiased finite-sample estimator. Nevertheless M is complete sufficient for this shift family under the usual integrability conditions, and $M - 1/n$ is its UMVU estimator. Completeness follows from $E_\theta h(M) = \int_\theta^\infty h(m) n e^{-n(m-\theta)} dm = 0$: differentiating the zero tail integral in θ gives $h(\theta) = 0$ almost everywhere. The MLE has MSE $2/n^2$, the bias-corrected estimator has MSE $1/n^2$, and both are consistent at rate n. There is no regular Gaussian $\sqrt{n}$ efficiency comparison here; using that term requires a specified non-regular optimality criterion.

Chapter 6

4 February 2013

6.1 Question 1

Enunciato. Describe hypothesis testing according to the Neyman–Pearson lemma. Let $(X_1, \dots, X_n)$ have density $f_X(x;\theta) = \exp\{A(x)Q(\theta) + C(x) - K(\theta)\}\mathbf{1}_S(x)$, where $S \subset \mathbb{R}$ and A, Q, C, K are known. If Q is increasing, what is the shape of the rejection region for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$?

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0,1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x)\exp\{\eta(\theta)T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \mathrm{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \leq \alpha \leq P_{\theta_0}(T \geq k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t}\theta^t(1-\theta)^{n-t} + \gamma\binom{n}{k}\theta^k(1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

For this model the log likelihood ratio is $[Q(\theta_1) - Q(\theta_0)]\sum A(X_i) - n[K(\theta_1) - K(\theta_0)]$. Its coefficient of the sample sum is nonnegative and positive when Q is strictly increasing at the two parameters. The MP region is $\sum A(X_i) > c$, with randomization at equality to attain α. Choose c from the null distribution of the sum. If $Q(\theta_1) = Q(\theta_0)$ and normalization makes the two laws equal, every exact-size test has the same power.

6.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ have density $f_X(x;\theta) = 4x^2 e^{-x^2/\theta^2}/(\theta^3\sqrt{\pi})\mathbf{1}_{\mathbb{R}^+}(x)$, $\theta > 0$. Find a sufficient statistic and the MLE. Compute the second moment of the MLE; can you infer it is unbiased? Compute the first moment of X_1; find an unbiased estimator of θ and its mean square error.

Soluzione. The normalized model is $f_\theta(x) = 4x^2 e^{-x^2/\theta^2}/(\theta^3\sqrt{\pi})$ on $x > 0$. Its sample likelihood is $C(x)\theta^{-3n}\exp[-T/\theta^2]$, $T = \sum X_i^2$, so T is sufficient by factorization. The score $-3n/\theta + 2T/\theta^3$ vanishes at

$$\hat{\theta}_{ML} = \sqrt{\frac{2T}{3n}}.$$

It is the global maximum because the likelihood vanishes at both ends. Transforming $W = X^2/\theta^2$ gives Gamma$(3/2, 1)$, hence $T/\theta^2 \sim$ Gamma$(3n/2, 1)$. Therefore $E\hat{\theta}^2_{ML} = \theta^2$, but this does not imply unbiasedness; strict Jensen gives $E\hat{\theta}_{ML} < \theta$. Explicitly

$$E\hat{\theta}_{ML} = \theta\sqrt{\frac{2}{3n}}\frac{\Gamma((3n+1)/2)}{\Gamma(3n/2)}.$$

Gamma integration also gives $EX = 2\theta/\sqrt{\pi}$, $EX^2 = 3\theta^2/2$. Thus $U = (\sqrt{\pi}/2)\bar{X}$ is unbiased, and its MSE equals its variance:

$$\mathrm{MSE}(U) = \frac{\theta^2}{n}\left(\frac{3\pi}{8} - 1\right).$$

The sample second moment consistently estimates $3\theta^2/2$, so the MLE is strongly consistent; its bias vanishes. An exactly unbiased version of the MLE can also be formed by dividing by the explicit Gamma-ratio factor.

Chapter 7

10 June 2013

7.1 Question 1

Enunciato. Introduce the likelihood function and describe maximum likelihood point estimation. Let $(X_1, \dots, X_n)$ have density $f_X(x;\theta) = 4x^2 e^{-x^2/\theta^2}/(\theta^3\sqrt{\pi})\mathbf{1}_{\mathbb{R}^+}(x)$, $\theta > 0$. Find a sufficient statistic and the MLE of θ. Provide and solve at least one point estimation problem in which the usual likelihood regularity conditions are not satisfied.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \le 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat\theta_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat\theta_n - \theta_0), \quad \sqrt{n}(\hat\theta_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat\theta_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \mathrm{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log\theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat\theta = 1/\bar X, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat\theta \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat\theta - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat\theta = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat\mu = \bar X$ and $\hat\sigma^2 = n^{-1}\sum (X_i - \bar X)^2$; the latter has expectation $(n-1)\sigma^2/n$.

The normalized model is $f_\theta(x) = 4x^2 e^{-x^2/\theta^2}/(\theta^3\sqrt{\pi})$ on $x > 0$. Its sample likelihood is $C(x)\theta^{-3n}\exp[-T/\theta^2]$, $T = \sum X_i^2$, so T is sufficient by factorization. The score $-3n/\theta + 2T/\theta^3$ vanishes at

$$\hat{\theta}_{ML} = \sqrt{\frac{2T}{3n}}.$$

It is the global maximum because the likelihood vanishes at both ends. Transforming $W = X^2/\theta^2$ gives Gamma$(3/2, 1)$, hence $T/\theta^2 \sim$ Gamma$(3n/2, 1)$. Therefore $E\hat{\theta}_{ML}^2 = \theta^2$, but this does not imply unbiasedness; strict Jensen gives $E\hat{\theta}_{ML} < \theta$. Explicitly

$$E\hat{\theta}_{ML} = \theta\sqrt{\frac{2}{3n}}\frac{\Gamma((3n+1)/2)}{\Gamma(3n/2)}.$$

Gamma integration also gives $EX = 2\theta/\sqrt{\pi}$, $EX^2 = 3\theta^2/2$. Thus $U = (\sqrt{\pi}/2)\bar{X}$ is unbiased, and its MSE equals its variance:

$$\mathrm{MSE}(U) = \frac{\theta^2}{n}\left(\frac{3\pi}{8} - 1\right).$$

The sample second moment consistently estimates $3\theta^2/2$, so the MLE is strongly consistent; its bias vanishes. An exactly unbiased version of the MLE can also be formed by dividing by the explicit Gamma-ratio factor.

Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0, \theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat{\theta} = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \mathrm{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0, 1)$ since $P((M/\theta_0)^n \le u) = u$. Consequently $-2\log\Lambda$ has the χ_2^2 distribution, not Wilks' predicted χ_1^2. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M - \theta) \sim \mathrm{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ_2^2. These examples show why calibrating automatically by χ_1^2 is incorrect. For $U(a - b, a + b)$ with known b, admissible centers lie between $M - b$ and $m + b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

7.2 Question 2

Enunciato. The central unit of a computer receives service requests from users. Every service binds the memory for an exponential time with known parameter λ. If the central unit was active for total time $X = x$, what is the maximum likelihood estimator of the number n of requests? Suppose $\lambda = 3$ and $x = 227$; what is the estimate?

Soluzione. Assume independent service durations and interpret the known exponential parameter as rate. For an integer number $n \geq 1$ of sequential services, $X \mid n \sim \text{Gamma}(n, \lambda)$, giving $L(n) = \lambda^n x^{n-1} e^{-\lambda x}/(n-1)!$. Its successive ratio is $L(n+1)/L(n) = \lambda x/n$. If λx is noninteger, the unique maximum is $\lceil \lambda x \rceil$. If $\lambda x = m$ is an integer, both m and $m+1$ maximize. For $\lambda = 3, x = 227$, $m = 681$, so 681 and 682 are tied MLEs. If λ were a scale parameter instead, replace λx by x/λ throughout; the convention must be stated.

Chapter 8

2 February 2017

8.1 Question 1(i)

Enunciato. Let $X = (X_1, \dots, X_n)$ be a random sample from $\{f_\theta : \theta \in \Theta \subset \mathbb{R}\}$. Describe efficiency for an estimator of θ. State and prove a necessary and sufficient condition for existence of an efficient unbiased estimator.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1 - \theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

8.2 Question 1(ii)

Enunciato. Let $X = (X_1, \dots, X_n)$ be a random sample from $\{f_\theta : \theta \in \Theta \subset \mathbb{R}\}$. Let ψ be defined on Θ and T be an unbiased estimator of $\psi(\theta)$ with $E_\theta T^2 < \infty$ for all parameters. Assume distinct parameters give distinct distributions and that $S(\varphi) = \{f_\varphi > 0\} \subset S(\theta) = \{f_\theta > 0\}$ for some $\varphi \neq \theta$. Show that

$$\mathrm{Var}_\theta\, T \geq \sup_{\varphi \neq \theta : S(\varphi) \subset S(\theta)} \frac{[\psi(\varphi) - \psi(\theta)]^2}{\mathrm{Var}_\theta[f_\varphi(X)/f_\theta(X)]}.$$

Soluzione. Let $R_\varphi(X) = f_\varphi(X)/f_\theta(X)$ on $S(\theta)$. The support inclusion ensures $E_\theta R_\varphi = \int f_\varphi = 1$ and $E_\theta[T R_\varphi] = E_\varphi T = \psi(\varphi)$. Since $E_\theta T = \psi(\theta)$,

$$\mathrm{Cov}_\theta(T, R_\varphi) = \psi(\varphi) - \psi(\theta).$$

Cauchy–Schwarz gives $\mathrm{Var}_\theta T \geq [\psi(\varphi) - \psi(\theta)]^2 / \mathrm{Var}_\theta R_\varphi$ for every admissible $\varphi \neq \theta$ with finite positive denominator. Taking the supremum proves the Hammersley–Chapman–Robbins bound. If the denominator is infinite the corresponding bound is interpreted as zero; if it is zero the distributions coincide and the numerator is zero, so that term must be excluded rather than dividing $0/0$. No likelihood derivatives or common support equalities are required, making this bound useful in non-regular models.

8.3 Question 2

Enunciato. Let $(X_1, \ldots, X_n)$ have density $f_\theta(x) = \theta^{-1} x^{(1-\theta)/\theta} \mathbf{1}_{(0,1)}(x)$, $\theta > 0$. (i) Find, if it exists, a UMP size-α test of $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$. (ii) For $n = 2$, $\theta_0 = 1$, $\alpha = 0.05$, represent the power function.

Soluzione. Put $Z_i = -\log X_i$. Then $P(Z_i > z) = e^{-z/\theta}$, so Z_i is exponential with scale θ and $T = \sum Z_i \sim \mathrm{Gamma}(n, \text{scale } \theta)$. The ratio for $\theta_1 > \theta_0$ is $(\theta_0/\theta_1)^n \exp[T(1/\theta_0 - 1/\theta_1)]$, increasing in T. Reject $T > \theta_0 G_{n,1-\alpha}$, where G is a unit-scale Gamma quantile. Stochastic monotonicity calibrates the composite null at its largest parameter θ_0, and NP gives UMP against every larger scale. For $n = 2$, $P_\theta(T > c) = e^{-c/\theta}(1 + c/\theta)$. Choose c as the unique positive solution $e^{-c}(1 + c) = 0.05$ ($c \simeq 4.743865$). The full power function is $\beta(\theta) = e^{-c/\theta}(1 + c/\theta)$, increasing from zero to one and equal to 0.05 at $\theta = 1$.

Chapter 9

20 February 2019

9.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $\{f_\theta : \theta \in \Theta \subset \mathbb{R}\}$. (i) Describe efficiency and state and prove a necessary and sufficient condition for an efficient unbiased estimator of θ. (ii) Prove that the exponential family is a parametric family for which there exists an efficient estimator, biased or unbiased, of the parameter of interest.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1 - \theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

Write a one-parameter exponential family as $f_\theta(x) = h(x) \exp\{\eta(\theta)A(x) - \psi(\theta)\}$ on fixed support. In the natural parameter $t = \eta(\theta)$ the log partition is $a(t) = \log \int h(x) e^{tA(x)} dx$. Differentiating under the integral gives

$$a'(t) = E_t A(X), \quad a''(t) = \mathrm{Var}_t\, A(X) \geq 0.$$

The sample likelihood depends on data only through $T = \sum A(X_i)$, proving sufficiency by factorization. Its natural score is $T - na'(t)$, information is $na''(t)$ and the likelihood equation is $a'(\hat{t}) = T/n$. Strict positive variance makes a strictly convex and the log likelihood strictly concave, hence at most one

interior maximum. A boundary maximum may still occur. Bernoulli has $A = x$, $t = \log[\theta/(1-\theta)]$, $a = \log(1+e^t)$, giving mean θ and variance $\theta(1-\theta)$. Poisson has $A = x$, $t = \log\lambda$, $a = e^t$, giving both mean and variance λ.

Part (ii) requires identifying the target. In a natural family the sample average of A is an unbiased efficient estimator of its mean $m = a'(t)$: its variance is $a''(t)/n$ and the functional CR bound is $[m'(t)]^2/[na''(t)] = a''(t)/n$. If the requested parameter is an arbitrary nonlinear parametrization, exact unbiased efficiency is not automatic. For example Gamma shape fixed with unknown rate has a biased MLE of the rate. Its sample mean is efficient for the inverse-rate mean functional, not the rate itself. The generally valid claim is efficiency of the mean sufficient-statistic parameter and asymptotic efficiency of regular interior MLEs, not universal finite-sample efficiency for every parameter.

9.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ have density

$$f_\theta(x) = \begin{cases} 2x/(\alpha\theta), & 0 \le x \le \theta, \\ 2(\alpha - x)/[\alpha(\alpha-\theta)], & \theta \le x \le \alpha, \\ 0, & \text{otherwise}, \end{cases}$$

where $\alpha > 0$ is known and $0 < \theta < \alpha$. Find the MLE of θ. Compute it for $n = 3$, $(x_1, x_2, x_3) = (1, 2, 4)$ and $\alpha = 5$.

Soluzione. Sort the data. On an interval between consecutive observations there are j observations below θ, so $\ell(\theta) = C - j\log\theta - (n-j)\log(\alpha-\theta)$. Its second derivative is $j/\theta^2 + (n-j)/(\alpha-\theta)^2 > 0$, so any stationary point inside that interval is a minimum. Maxima must occur at observed knots or parameter-space boundaries. For data strictly inside $(0, \alpha)$, likelihood increases on $(0, x_{(1)})$ and decreases on $(x_{(n)}, \alpha)$, eliminating the outer limits; evaluate it at each distinct observation.

For $(1, 2, 4)$ and $\alpha = 5$, the density products at $\theta = 1, 2, 4$ are respectively $3/100$, $2/75$, and $1/50$. The largest is $3/100$, so $\hat\theta = 1$. Ties may give multiple MLEs in other samples. Observations at the endpoints have probability zero and require separate density-version conventions.

Chapter 10

24 February 2019

10.1 Question 1

Enunciato. Illustrate the theory of statistical hypothesis testing according to the likelihood ratio test. Let $\bar{X} = (X_1, \dots, X_n)$ be a random sample from $N(\mu_x, \sigma_x^2)$ and $\bar{Y} = (Y_1, \dots, Y_m)$ an independent sample from $N(\mu_y, \sigma_y^2)$. Describe the shape and distribution of the rejection region for $H_0 : \sigma_x^2 = \sigma_y^2$ versus $H_1 : \sigma_x^2 \neq \sigma_y^2$.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}, \qquad D = -2 \log \Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta \phi$, size $\sup_{\Theta_0} \beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi_r^2$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat{\theta}$ gives

$$D = -\ell_n''(\tilde{\theta})(\hat{\theta} - \theta_0)^2 = \left[-\ell_n''(\tilde{\theta})/n\right] \left[\sqrt{n}(\hat{\theta} - \theta_0)\right]^2 \Rightarrow \chi_1^2.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi_1^2$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ_r^2. Reject when $D > \chi_{r,1-\alpha}^2$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat{\lambda} = \bar{X}$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar{X} \log(\bar{X}/\lambda_0) - (\bar{X} - \lambda_0)\},$$

with $0 \log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar{X} - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ_1^2 limit. Wald uses $n(\bar{X} - \lambda_0)^2/\bar{X}$ (when $\bar{X} > 0$), score uses $n(\bar{X} - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

Let two independent normal samples have means μ_x, μ_y and sample sizes n, m. For a common unknown variance, set $R_x = \sum(X_i - \bar{X})^2$, $R_y = \sum(Y_j - \bar{Y})^2$ and $s_p^2 = (R_x + R_y)/(n + m - 2)$. Under equal means the MLE of the common mean is $(n\bar{X} + m\bar{Y})/(n + m)$; the restricted residual sum is

$$R_0 = R_x + R_y + \frac{nm}{n + m}(\bar{X} - \bar{Y})^2.$$

Under the alternative the two fitted means are the sample means. Maximizing variance gives

$$\Lambda = \left(\frac{R_x + R_y}{R_0}\right)^{(n+m)/2} = \left(1 + \frac{T^2}{n+m-2}\right)^{-(n+m)/2}, \quad T = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{1/n + 1/m}}.$$

Under H_0, $T \sim t_{n+m-2}$, because the normalized mean difference is standard Gaussian and independent of the pooled chi-square residual. Reject $|T| > t_{n+m-2,1-\alpha/2}$.

For testing equal variances with means unrestricted, the unrestricted variance MLEs are $R_x/n, R_y/m$; the restricted estimate is $(R_x + R_y)/(n+m)$. Thus

$$\Lambda = \frac{(R_x/n)^{n/2}(R_y/m)^{m/2}}{[(R_x + R_y)/(n+m)]^{(n+m)/2}}.$$

It is largest at $R_x/n = R_y/m$ and decreases in either direction. The statistic $F = (R_x/(n-1))/(R_y/(m-1))$ has $F_{n-1,m-1}$ null law. The LRT rejects for $F < a$ or $F > b$, where $\Lambda(a) = \Lambda(b)$ and $P(F < a) + P(F > b) = \alpha$. The common equal-tail F test uses quantiles $F_{\alpha/2}, F_{1-\alpha/2}$ and has exact size, but is not generally the same as the equal-likelihood LRT cutoffs. Without normality the exact t/F pivots fail: with finite variances the two-sample CLT and estimated standard errors yield a large-sample normal mean test; variance tests require finite fourth moments and a fourth-moment-adjusted standard error. Gaussian likelihood procedures under misspecification require sandwich covariance, not an automatic Wilks assertion.

10.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from a Poisson random variable of parameter $\lambda > 0$, and consider estimating $\theta = e^{-\lambda}$. For $n = 1$, show that $T_1 = \delta_{X_1}$ is the unique unbiased minimum-variance estimator of θ. For arbitrary $n \geq 1$, is $T_n = n^{-1}\sum_i \delta_{X_i}$ unbiased and efficient?

Soluzione. Here $\delta_x = \mathbf{1}_{\{x=0\}}$, so $E\delta_X = P(X = 0) = e^{-\lambda} = \theta$. Poisson sampling is complete: if $Eh(X) = 0$ for every $\lambda > 0$ with integrability, then $\sum_{k\geq 0} h(k)\lambda^k/k! = 0$; uniqueness of the power series gives every $h(k) = 0$. For $n = 1$ the full observation is sufficient and complete; two unbiased estimators must coincide pointwise, so δ_X is uniquely unbiased and UMVU.

For general n, independence gives $ET_n = \theta$ and $\operatorname{Var} T_n = \theta(1-\theta)/n$. Information for λ is n/λ; for the functional $e^{-\lambda}$ the Cramér–Rao bound is $\lambda e^{-2\lambda}/n$. Since $e^{\lambda} - 1 > \lambda$, T_n does not attain this bound (including $n = 1$); being UMVU need not mean CR efficient. For $n > 1$ it is not UMVU either: with $S = \sum X_i \sim \text{Pois}(n\lambda)$, conditioning gives

$$E[T_n \mid S] = (1 - 1/n)^S,$$

because $X_i \mid S \sim \text{Bin}(S, 1/n)$. Its expectation is $\exp\{n\lambda[(1-1/n) - 1]\} = e^{-\lambda}$ and its variance is $e^{-2\lambda}(e^{\lambda/n} - 1)$, strictly smaller for $n > 1$. The complete sufficient statistic S makes this the UMVU estimator. Its asymptotic variance is $\lambda e^{-2\lambda}/n$, attaining the regular asymptotic bound.

Chapter 11

11 June 2019

11.1 Question 1

Enunciato. Let $Y = X\beta + \varepsilon$, where $Y = (Y_1, \dots, Y_n)^T$ is the response, $X = (x_{ij})$ is the $n \times p$ design matrix with $n \geq p$, ε is the error vector and $\beta \in \mathbb{R}^p$. Describe least squares estimation of β and show it coincides with MLE under Gaussian errors. Using the likelihood ratio approach, propose a statistic and its distribution for $H_0 : \beta_r = 0$ versus $H_1 : \beta_r \neq 0$, $1 \leq r \leq p$.

Soluzione. Assume full column rank, $n > p$ and independent errors $N(0, \sigma^2)$ with unknown common variance. Least squares minimizes $\|y - X\beta\|^2$; the normal equations $X^T X\hat{\beta} = X^T y$ give $\hat{\beta} = (X^T X)^{-1} X^T y$. The Gaussian log likelihood is $-n \log \sigma^2/2 - \|y - X\beta\|^2/(2\sigma^2)$ plus a constant, so profiling yields the same coefficient estimate and $\hat{\sigma}^2 = RSS_1/n$.

Under $\beta_r = 0$ minimize residual sum subject to that restriction to obtain RSS_0. Orthogonal projections give $(RSS_0 - RSS_1)/\sigma^2 \sim \chi^2_1$, $RSS_1/\sigma^2 \sim \chi^2_{n-p}$ independently under the null. The likelihood ratio is $(RSS_1/RSS_0)^{n/2}$, decreasing in

$$F = \frac{RSS_0 - RSS_1}{RSS_1/(n-p)} \sim F_{1,n-p}.$$

Reject $F > F_{1,n-p,1-\alpha}$. Equivalently $t = \hat{\beta}_r / [s\sqrt{(X^T X)^{-1}_{rr}}]$ has t_{n-p} law, $s^2 = RSS_1/(n-p)$, and reject $|t| > t_{n-p,1-\alpha/2}$. With known variance use $\hat{\beta}_r / [\sigma\sqrt{(X^T X)^{-1}_{rr}}] \sim N(0,1)$. For rank-deficient design coefficients need not be identifiable; $n = p$ leaves no residual degrees of freedom for this unknown-variance test.

11.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from a Poisson random variable of parameter $\lambda > 0$, and consider estimating $\theta = e^{-\lambda}$. For $n = 1$, show that $T_1 = \delta_{X_1}$, where $\delta_x = 1$ for $x = 0$ and zero for $x \geq 1$ is the unique unbiased minimum-variance estimator of θ. For arbitrary $n \geq 1$, is $T_n = n^{-1} \sum_i \delta_{X_i}$ unbiased and efficient?

Soluzione. Here $\delta_x = \mathbf{1}_{\{x=0\}}$, so $E\delta_X = P(X = 0) = e^{-\lambda} = \theta$. Poisson sampling is complete: if $Eh(X) = 0$ for every $\lambda > 0$ with integrability, then $\sum_{k \geq 0} h(k)\lambda^k/k! = 0$; uniqueness of the power series gives every $h(k) = 0$. For $n = 1$ the full observation is sufficient and complete; two unbiased estimators must coincide pointwise, so δ_X is uniquely unbiased and UMVU.

For general n, independence gives $ET_n = \theta$ and $\operatorname{Var} T_n = \theta(1-\theta)/n$. Information for λ is n/λ; for the functional $e^{-\lambda}$ the Cramér–Rao bound is $\lambda e^{-2\lambda}/n$. Since $e^{\lambda} - 1 > \lambda$, T_n does not attain this bound (including $n = 1$); being UMVU need not mean CR efficient. For $n > 1$ it is not UMVU either:

with $S = \sum X_i \sim \text{Pois}(n\lambda)$, conditioning gives

$$E[T_n \mid S] = (1 - 1/n)^S,$$

because $X_i \mid S \sim \text{Bin}(S, 1/n)$. Its expectation is $\exp\{n\lambda[(1 - 1/n) - 1]\} = e^{-\lambda}$ and its variance is $e^{-2\lambda}(e^{\lambda/n} - 1)$, strictly smaller for $n > 1$. The complete sufficient statistic S makes this the UMVU estimator. Its asymptotic variance is $\lambda e^{-2\lambda}/n$, attaining the regular asymptotic bound.

Chapter 12

January 2020

12.1 Exercise 1(i)

Enunciato. Let $X = (X_1, \dots, X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta \subset \mathbb{R}\}$. Define efficiency, both asymptotic and non-asymptotic.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1 - \theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

12.2 Exercise 1(ii)

Enunciato. Let $X = (X_1, \dots, X_n)$ be a random sample from $\{f_\theta : \theta \in \Theta \subset \mathbb{R}\}$. Let ψ be defined on Θ and T be an unbiased estimator of $\psi(\theta)$ with $E_\theta T^2 < \infty$ for all parameters. Assume distinct parameters give distinct distributions and that $S(\varphi) = \{f_\varphi > 0\} \subset S(\theta) = \{f_\theta > 0\}$ for some $\varphi \neq \theta$. Show that

$$\mathrm{Var}_\theta\, T \geq \sup_{\varphi \neq \theta : S(\varphi) \subset S(\theta)} \frac{[\psi(\varphi) - \psi(\theta)]^2}{\mathrm{Var}_\theta[f_\varphi(X)/f_\theta(X)]}.$$

Soluzione. Let $R_\varphi(X) = f_\varphi(X)/f_\theta(X)$ on $S(\theta)$. The support inclusion ensures $E_\theta R_\varphi = \int f_\varphi = 1$ and $E_\theta[TR_\varphi] = E_\varphi T = \psi(\varphi)$. Since $E_\theta T = \psi(\theta)$,

$$\operatorname{Cov}_\theta(T, R_\varphi) = \psi(\varphi) - \psi(\theta).$$

Cauchy–Schwarz gives $\operatorname{Var}_\theta T \geq [\psi(\varphi) - \psi(\theta)]^2 / \operatorname{Var}_\theta R_\varphi$ for every admissible $\varphi \neq \theta$ with finite positive denominator. Taking the supremum proves the Hammersley–Chapman–Robbins bound. If the denominator is infinite the corresponding bound is interpreted as zero; if it is zero the distributions coincide and the numerator is zero, so that term must be excluded rather than dividing $0/0$. No likelihood derivatives or common support equalities are required, making this bound useful in non-regular models.

12.3 Exercise 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $\{f_\mu : \mu \in M \subset \mathbb{R}\}$ with $EX_1 = \mu$, $\operatorname{Var} X_1 = 1$, $EX_1^4 < \infty$. The unbiased estimators of μ^2 are $T_{1,n} = n^{-1}\sum X_i^2 - 1$ and $T_{2,n} = \bar{X}_n^2 - 1/n$. (i) Find $\lim_n \operatorname{eff}(T_{1,n}, T_{2,n})$. (ii) Show that it is at most one if $X_i - \mu$ is symmetric. (iii) Find a parametric family for which it exceeds one. The relative efficiency is $E[(T_2 - \theta)^2]/E[(T_1 - \theta)^2]$. You may use the Delta method and vanishing odd central moments of symmetric distributions.

Soluzione. Write $Z = X - \mu$, $EZ = 0$, $EZ^2 = 1$, $\kappa_3 = EZ^3$, $\kappa_4 = EZ^4$. Expansion gives

$$v_1 = \operatorname{Var}(X^2) = 4\mu^2 + 4\mu\kappa_3 + \kappa_4 - 1, \quad \operatorname{Var} T_{1,n} = v_1/n.$$

The second estimator is unbiased since $E\bar{X}^2 = \mu^2 + 1/n$. Expanding $\bar{X} = \mu + \bar{Z}$ and using $E\bar{Z}^3 = \kappa_3/n^2$, $E\bar{Z}^4 = 3/n^2 + (\kappa_4 - 3)/n^3$ gives the exact expression

$$\operatorname{Var} T_{2,n} = \frac{4\mu^2}{n} + \frac{4\mu\kappa_3 + 2}{n^2} + \frac{\kappa_4 - 3}{n^3}.$$

Thus, if $v_1 > 0$, the relative efficiency limit is $4\mu^2/v_1$; this also gives zero at $\mu = 0$. If the centered law is symmetric, $\kappa_3 = 0$ and $\kappa_4 \geq (EZ^2)^2 = 1$, so $v_1 \geq 4\mu^2$, proving the bound. Degenerate cases $v_1 = 0$ must be considered separately: the first estimator may have zero risk and the ratio is undefined rather than assigned a finite value.

For a counterexample with a fixed-shape location family, take $X = \mu + Z$, where $Z = -2$ with probability $1/5$ and $Z = 1/2$ with probability $4/5$. Then $EZ = 0$, $EZ^2 = 1$, $\kappa_3 = -3/2$ and $\kappa_4 = 13/4$. At $\mu = 1$, $v_1 = 4 - 6 + 9/4 = 1/4$ and the limit equals $16 > 1$. All fourth moments are finite.

Chapter 13

February 2020 – handwritten recollection

13.1 Question 1

Enunciato. Speak about the likelihood ratio test for general hypotheses, including its asymptotic and non-asymptotic theory.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

For iid $N(\mu, \sigma^2)$ with both parameters unknown, the unrestricted MLEs are $\bar X$ and R/n, $R = \sum(X_i - \bar X)^2$. Under $\mu = \mu_0$, the variance MLE is $[R + n(\bar X - \mu_0)^2]/n$. Maximized Gaussian likelihood therefore gives

$$\Lambda = \left(1 + \frac{n(\bar X - \mu_0)^2}{R}\right)^{-n/2} = \left(1 + \frac{T^2}{n-1}\right)^{-n/2}, \quad T = \frac{\sqrt{n}(\bar X - \mu_0)}{S},\ S^2 = R/(n-1).$$

Normal orthogonal projections give $\sqrt{n}(\bar{X}-\mu_0)/\sigma \sim N(0,1)$ independently of $R/\sigma^2 \sim \chi^2_{n-1}$, so $T \sim t_{n-1}$. The exact level-α region is $|T| > t_{n-1,1-\alpha/2}$.

Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0,\theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat{\theta} = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1-M/\theta) > t) = (1-t/n)^n \to e^{-t}.$$

Thus $n(\theta-M)/\theta \Rightarrow \mathrm{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \le u) = u$. Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \mathrm{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b,a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

13.2 Question 2

Enunciato. Show that, for a parametric family $F = \{f_\theta(x) : \theta \in \Theta\}$ with positive smooth densities, the monotone likelihood ratio property is equivalent to either of the conditions

$$\frac{\partial^2}{\partial\theta\partial x}\log f_\theta(x) \ge 0, \qquad f_\theta(x)\frac{\partial^2 f_\theta(x)}{\partial\theta\partial x} \ge \frac{\partial f_\theta(x)}{\partial\theta}\frac{\partial f_\theta(x)}{\partial x}.$$

Soluzione. Assume a common interval support, an interval parameter set and twice continuously differentiable strictly positive density. MLR means $f_{\theta_1}(x)/f_{\theta_0}(x)$ is nondecreasing in x for every $\theta_1 > \theta_0$. Taking logarithms and differentiating gives $\partial_x \log f_{\theta_1} - \partial_x \log f_{\theta_0} \ge 0$. Thus $\partial_x \log f_\theta$ is nondecreasing in θ, equivalent by smoothness to its θ derivative being nonnegative. Conversely integrating the cross derivative in θ proves the likelihood-ratio derivative is nonnegative. Finally

$$\partial^2_{\theta x}\log f = \frac{f f_{\theta x} - f_\theta f_x}{f^2};$$

the positive denominator gives the second criterion. If the handwritten version uses strict $>$, that is sufficient for strictly increasing ratios but is stronger than ordinary weak MLR; nonnegative derivatives are the exact weak equivalence.

13.3 Question 3

Enunciato. Let p be a parametric family of densities of a noncentral chi-square distribution with noncentrality parameter $\theta > 0$ and a fixed number of degrees of freedom. Show that it has the monotone likelihood ratio property.

Soluzione. Fix $\nu > 0$. The noncentral chi-square density is the Poisson mixture

$$f_\theta(x) = e^{-\theta/2} \sum_{j=0}^{\infty} \frac{(\theta/2)^j}{j!} \frac{x^{\nu/2+j-1} e^{-x/2}}{2^{\nu/2+j}\Gamma(\nu/2+j)}, \quad x > 0.$$

Factor out $x^{\nu/2-1}e^{-(x+\theta)/2}$ and write the remaining series as $H(t) = \sum_{j\geq 0} c_j t^j$ with $t = \theta x$ and positive $c_j = 1/[2^{\nu/2+2j} j!\Gamma(\nu/2+j)]$. The series and its derivatives converge locally uniformly. Introduce a distribution on integers with probabilities $P_t(J = j) = c_j t^j / H(t)$. Then $\partial_\theta \log f_\theta(x) = -1/2 + E_t J/\theta$, and differentiating with respect to x gives

$$\partial^2_{\theta x} \log f_\theta(x) = \frac{\operatorname{Var}_t J}{\theta x} > 0.$$

Indeed differentiation with respect to $\log t$ turns a weighted mean into the corresponding variance. The positivity follows from at least two nonzero series coefficients. The cross-derivative criterion now proves MLR in the observation x. For a single observation this yields an upper-tail UMP test for larger noncentrality; the density's Poisson-mixture construction supplies the exact null quantiles.

13.4 Question 4

Enunciato. Let $(X_1, \ldots, X_n)$ be a random sample from $f_\theta(x) = \theta x^{-2} \mathbf{1}_{(\theta,+\infty)}(x)$. Find a UMP size-$\alpha$ test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.

Soluzione. Normalization requires $\theta > 0$. Put $M = \min X_i$. The likelihood is $\theta^n \prod x_i^{-2} \mathbf{1}_{\{M>\theta\}}$. For $\theta_1 > \theta_0$ the ratio is $(\theta_1/\theta_0)^n$ on $M > \theta_1$, zero on $\theta_0 < M \leq \theta_1$. Under the null $P(M > c) = (\theta_0/c)^n$ for $c \geq \theta_0$. Rejecting $M > c$ with $c = \theta_0 \alpha^{-1/n}$ has exact size α and is MP: if $c \geq \theta_1$ it uses the constant positive-ratio region, and if $c < \theta_1$ it contains the full alternative support, giving power one.

For $\theta_1 < \theta_0$ the alternative-only region $M \leq \theta_0$ costs no null size and must be rejected. On $M > \theta_0$ the ratio is constant $(\theta_1/\theta_0)^n$, so reject there independently with probability α. Its alternative power is $1 - (1-\alpha)(\theta_1/\theta_0)^n$. Any exact-size allocation on the null support is equally MP for that lower alternative. If $\theta_1 = \theta_0$, hypotheses coincide and every size-α test has power α.

Chapter 14

5 June 2020

14.1 Question 1

Enunciato. Illustrate the theory of statistical hypothesis testing according to the Neyman and Pearson lemma and its likelihood ratio generalization. Let $(X_1, \dots, X_n)$ be a random sample from $X \sim N(\theta, \sigma^2)$ with σ^2 unknown. Describe the shape of the rejection region for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0, 1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x) \exp\{\eta(\theta)T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \mathrm{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \leq \alpha \leq P_{\theta_0}(T \geq k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t} \theta^t (1-\theta)^{n-t} + \gamma \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}, \qquad D = -2 \log \Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0,1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta-\theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta-\theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X-\lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X-\lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X-\lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

For iid $N(\mu,\sigma^2)$ with both parameters unknown, the unrestricted MLEs are $\bar X$ and R/n, $R = \sum(X_i-\bar X)^2$. Under $\mu=\mu_0$, the variance MLE is $[R+n(\bar X-\mu_0)^2]/n$. Maximized Gaussian likelihood therefore gives

$$\Lambda = \left(1+\frac{n(\bar X-\mu_0)^2}{R}\right)^{-n/2} = \left(1+\frac{T^2}{n-1}\right)^{-n/2}, \quad T = \frac{\sqrt{n}(\bar X-\mu_0)}{S},\ S^2 = R/(n-1).$$

Normal orthogonal projections give $\sqrt{n}(\bar X-\mu_0)/\sigma \sim N(0,1)$ independently of $R/\sigma^2 \sim \chi^2_{n-1}$, so $T \sim t_{n-1}$. The exact level-α region is $|T| > t_{n-1,1-\alpha/2}$.

14.2 Question 2

Enunciato. A problem in surveys is that people tend to lie to protect their privacy. To overcome this drawback a common strategy is the following: we ask the interviewee to flip a coin that produces "YES" with probability p and "NO" with probability $1-p$, and then report if the outcome—which remains secret—coincides with the answer to the survey's question. It is assumed that the interviewee does not lie to this question. Suppose that the aim is to estimate the proportion θ of individuals in a certain group: which is the probability that the reported answer is "YES"? If Y_n is the number of interviewees reporting "YES", what is its distribution? What is an unbiased estimator of θ, its variance, and is it consistent?

Soluzione. Assume independent representative interviewees and independent coin flips. A coincidence occurs for a group member with a YES flip or a nonmember with a NO flip, so $q = p\theta+(1-p)(1-\theta) = 1-p+(2p-1)\theta$. Thus $Y_n \sim \text{Bin}(n,q)$. For $p \neq 1/2$,

$$\hat\theta = \frac{Y_n/n-(1-p)}{2p-1}, \qquad E\hat\theta = \theta, \quad \text{Var}\,\hat\theta = \frac{q(1-q)}{n(2p-1)^2}.$$

Variance tends to zero, giving mean-square and probability consistency; the strong LLN gives strong consistency. At $p = 1/2$, $q = 1/2$ for every θ, so the model is unidentifiable and no consistent estimator based on these responses can exist. The unbiased affine estimator may fall outside $[0,1]$ at finite samples; clipping produces a constrained MLE but generally destroys exact unbiasedness.

Chapter 15

21 December 2020 – slot 1

15.1 Question 1

Enunciato. Illustrate the theory of statistical hypothesis testing according to the likelihood ratio test.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X\log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

15.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $f_X(x;\theta) = \exp\{-(x-\theta)\}\mathbf{1}_{(\theta,+\infty)}(x)$. Find the maximum likelihood estimator of θ, verify whether it is unbiased for a finite sample size and asymptotically, and find an optimal rejection region of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.

Soluzione. Put $M = X_{(1)}$. The likelihood is $L(\theta) = e^{-\sum x_i + n\theta}\mathbf{1}_{\{\theta<M\}}$. It increases throughout its admissible parameter set. With the literal strict support it has no attained maximum; its supremum is approached as $\theta \uparrow M$. The conventional endpoint MLE, using an equivalent density version with $x \geq \theta$, is $\hat{\theta} = M$. Writing $X_i = \theta + E_i$, where E_i are iid Exp(1), gives $M - \theta \sim \text{Exp}(n)$. Hence

$$E\hat{\theta} = \theta + 1/n, \quad \text{Var}\,\hat{\theta} = 1/n^2, \quad \text{MSE}\,\hat{\theta} = 2/n^2.$$

It is biased at finite n, asymptotically unbiased and mean-square consistent. The estimator $M - 1/n$ is unbiased. For $H_0 : \theta = \theta_0$ versus $\theta_1 > \theta_0$, the likelihood ratio is zero if $M < \theta_1$ and constant $e^{n(\theta_1-\theta_0)}$ on $M \geq \theta_1$. Thus rejecting in a largest-M region is NP optimal. Choose

$$c = \theta_0 - \frac{\log \alpha}{n}, \qquad C = \{M \geq c\}.$$

Under the null $P(C) = e^{-n(c-\theta_0)} = \alpha$. Under $\theta > \theta_0$ its power is $e^{-n(c-\theta)}$ if $\theta < c$, and one if $\theta \geq c$. If $c < \theta_1$, the region contains the entire alternative support and has power one, so it is still most powerful. The same region is optimal for every positive shift, hence is UMP for the one-sided composite alternative.

Chapter 16

21 January 2021

16.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from the Uniform distribution on $(0, \theta)$, $\theta > 0$. Show that (i) $\frac{d}{d\theta} \int x f_\theta(x) dx \neq \int x \frac{d}{d\theta} f_\theta(x) dx$, where f_θ is the density of the largest order statistic $X_{(n)}$; (ii) the Cramér–Rao lower bound does not hold for $\hat{\theta} = (n+1)X_{(n)}/n$.

Soluzione. The maximum M has density $f_\theta(x) = nx^{n-1}\theta^{-n}\mathbf{1}_{(0,\theta)}(x)$ and $EM = n\theta/(n+1)$. Thus $d(EM)/d\theta = n/(n+1)$. Differentiating only the interior density gives

$$\int_0^\theta x \partial_\theta [nx^{n-1}\theta^{-n}] dx = -\frac{n}{\theta} EM = -\frac{n^2}{n+1},$$

which differs from $n/(n+1)$. The missing upper-boundary term is $\theta f_\theta(\theta) = n$. Leibniz' rule including that term recovers $n/(n+1)$. Also

$$EM^2 = \frac{n\theta^2}{n+2}, \quad \operatorname{Var} M = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

Therefore $\tilde{\theta} = (n+1)M/n$ is unbiased and has variance $\theta^2/[n(n+2)]$. Naively squaring the sample score $-n/\theta$ gives n^2/θ^2 and a supposed bound θ^2/n^2, larger than the actual variance; using expected negative second derivative even gives negative information. Neither is a valid regular information identity. The score is not centered and the moving support violates the Cramér–Rao assumptions, so there is no contradiction to the theorem.

16.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $N(\mu, 1)$ with unknown μ. Instead of recording all observations, one records only whether each observation is less than zero. Find the maximum likelihood estimator of μ based on the recorded information, and discuss its properties.

Soluzione. Let $K = \sum_i \mathbf{1}_{\{X_i<0\}}$. For $X_i \sim N(\mu, 1)$ the observed indicators are iid Bernoulli with success probability $q(\mu) = \Phi(-\mu)$. Their likelihood is $q^K(1-q)^{n-K}$, with score $K/q - (n-K)/(1-q)$ in the q parametrization. For $0 < K < n$, $\hat{q} = K/n$ and

$$\hat{\mu} = -\Phi^{-1}(K/n).$$

For $K = 0$ the supremum is approached as $\mu \to +\infty$, and for $K = n$ as $\mu \to -\infty$; no finite MLE exists in those cases. For any fixed finite true μ, the strong LLN gives $K/n \to \Phi(-\mu) \in (0, 1)$, proving eventual finiteness and strong consistency. The derivative of $-\Phi^{-1}(q)$ is $-1/\phi(\Phi^{-1}(q))$, so

$$\sqrt{n}(\hat{\mu} - \mu) \Rightarrow N\left(0, \frac{q(1-q)}{\phi(\mu)^2}\right).$$

The recorded sample loses information compared with the original measurements; the information per indicator is $\phi(\mu)^2/[q(1-q)]$.

Chapter 17

22 January 2021 – slot 1

17.1 Question 1

Enunciato. Introduce Fisher information and discuss its role within the problem of point estimation.

Soluzione. For an iid dominated model f_θ, the likelihood, log likelihood and score are

$$L_n(\theta; x) = \prod_i f_\theta(x_i), \quad \ell_n(\theta) = \sum_i \log f_\theta(x_i), \quad U_n(\theta) = \ell_n'(\theta).$$

Assume an open parameter space, common support, differentiability and an integrable domination permitting differentiation under the integral. Then

$$E_\theta U_1 = \int \partial_\theta f_\theta(x) dx = 0, \qquad I(\theta) = E_\theta U_1^2 = -E_\theta \partial_\theta^2 \log f_\theta(X).$$

Indeed $\partial_\theta^2 \log f = f''/f - (f'/f)^2$ and $\int f'' = 0$. Independence gives $I_n = nI$. Observed information is $J_n = -\ell_n''$, a random quantity, whereas expected information is its expectation. Information quantifies local distinguishability and the precision attainable by unbiased regular estimators. For Bernoulli observations, $U_n = (\sum X_i - n\theta)/[\theta(1-\theta)]$ and $I_n = n/[\theta(1-\theta)]$. Parameter-dependent support can invalidate differentiation and hence both identities.

An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

17.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample with density $f_X(x;\theta) = \frac{1}{2}e^{-|x-\theta|}\mathbf{1}_{\mathbb{R}}(x)$, $\theta \in \mathbb{R}$. Study sufficiency properties; find a method-of-moments estimator of θ; find the maximum likelihood estimator; does this model belong to the exponential family?

Soluzione. The sample likelihood is $2^{-n}\exp[-\sum|x_i-\theta|]$. The ordered sample is sufficient because the expression is symmetric, and in fact minimal sufficient: the likelihood ratio for two samples is parameter free iff $\sum|x_i-\theta| - \sum|y_i-\theta|$ is constant in θ. Its derivative away from observations is determined by their empirical counts; being identically zero implies equality of the multisets, including multiplicities. Thus all order statistics, not just the median, are needed in general.

Since $EX = \theta$, the moment estimator is $\bar{X}$, unbiased and consistent, with variance $2/n$. Maximizing likelihood minimizes $g(\theta) = \sum|x_i-\theta|$. Away from ties $g'(\theta) = \#\{x_i < \theta\} - \#\{x_i > \theta\}$. A minimizer has at least half the observations on either side: any sample median is an MLE. For odd n it is $x_{((n+1)/2)}$; for even n any point of $[x_{(n/2)}, x_{(n/2+1)}]$ maximizes. The location family is not a fixed finite-dimensional regular exponential family: the changing kink at $x = \theta$ prevents representing all $|x-\theta|$ by a fixed finite list of data functions. The full ordered sample's minimal sufficiency dimension grows with n, consistent with this conclusion.

Chapter 18

22 January 2021 – slot 2

18.1 Question 1

Enunciato. Describe the property of efficiency, finite and asymptotic, in the theory of point estimation based on the likelihood function.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T-\theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

18.2 Question 2

Enunciato. Let $(X_1, \ldots, X_n)$ be a random sample from $f_\theta(x) = \theta(1+x)^{-(1+\theta)} \mathbf{1}_{\mathbb{R}^+}(x)$, $\theta > 0$. Find a method-of-moments estimator $T_{1,n}$ of θ; find the maximum likelihood estimator $T_{2,n}$ of θ^{-1}; are these estimators unbiased?

Soluzione. This is a Lomax law with survival $(1+x)^{-\theta}$. For $\theta > 1$, tail integration gives $EX = \int_0^\infty (1+x)^{-\theta} dx = 1/(\theta - 1)$, hence $T_{1,n} = 1 + 1/\bar{X}$. For $0 < \theta \leq 1$ the first population moment is infinite, so that moment equation does not define a valid moment method. For $\theta > 1$, strict convexity of $1/x$ and nondegeneracy give $E[1/\bar{X}] > 1/E\bar{X} = \theta - 1$ whenever the expectation exists; if it diverges the estimator is certainly not unbiased. Thus $T_{1,n}$ is upward biased. At $n = 1$ its expectation is infinite because the density is positive near zero.

The log likelihood is $n \log \theta - (1+\theta) \sum \log(1 + X_i)$; differentiating gives $\hat{\theta} = n / \sum \log(1 + X_i)$. MLE invariance yields

$$T_{2,n} = \frac{1}{n} \sum_i \log(1 + X_i).$$

For $Z = \log(1 + X)$, $P(Z > z) = e^{-\theta z}$ for $z \geq 0$, so $Z \sim \text{Exp}(\theta)$. Therefore $ET_{2,n} = 1/\theta$ and $\operatorname{Var} T_{2,n} = 1/(n\theta^2)$: it is exactly unbiased and consistent.

Chapter 19

8 February 2021 – slot 1

19.1 Question 1

Enunciato. Consider a parametric family $F = \{f_\theta : \theta \in \Theta\}$ and describe unbiasedness and efficiency for an estimator T of θ.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T-\theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

19.2 Question 2

Enunciato. Let $f_X(x; \beta) = \beta^{-1} e^{-x/\beta} \mathbf{1}_{(0,+\infty)}(x)$, $\beta > 0$, and let $(X_1, \ldots, X_n)$ be a random sample. Find the maximum likelihood estimator of β when the sole information available is that $0 \leq k \leq n$ observations are less than or equal to a fixed positive number M.

Soluzione. Each recorded event has probability $q(\beta) = 1 - e^{-M/\beta}$, so $K \sim \mathrm{Bin}(n, q)$. Its log likelihood, apart from a constant, is $k \log q + (n-k)\log(1-q)$. If $0 < k < n$ the maximizing success

probability is k/n; inversion gives

$$\hat{\beta} = -\frac{M}{\log(1 - k/n)}.$$

For $k = 0$, the supremum is approached as $\beta \to \infty$; for $k = n$, as $\beta \downarrow 0$. Neither boundary value belongs to the parameter space, so no finite attained MLE exists there.

Chapter 20

9 February 2021

20.1 Question 1

Enunciato. Let X_{ij}, $j = 1, \dots, r > 1$ and $i = 1, \dots, n$, be independent Gaussian variables with parameters (μ_i, σ^2). Find the MLE of $\theta = (\mu_1, \dots, \mu_n, \sigma^2)$ and verify consistency, as $n \to \infty$, of the variance MLE.

Soluzione. The log likelihood is $-nr \log \sigma^2/2 - \sum_{i,j}(x_{ij} - \mu_i)^2/(2\sigma^2)$ plus a constant. For each i, differentiation gives $\hat{\mu}_i = \bar{X}_i = r^{-1} \sum_j X_{ij}$; profiling variance gives

$$\hat{\sigma}^2 = \frac{1}{nr} \sum_{i=1}^{n} \sum_{j=1}^{r} (X_{ij} - \bar{X}_i)^2.$$

Each independent group residual sum divided by σ^2 has χ^2_{r-1} law. Thus $E\hat{\sigma}^2 = (r-1)\sigma^2/r$ and $\operatorname{Var} \hat{\sigma}^2 = 2(r-1)\sigma^4/(nr^2) \to 0$. With fixed r, it converges to $(r-1)\sigma^2/r$, not σ^2. This is the incidental-parameter failure: the number of nuisance means grows with n. Multiplying by $r/(r-1)$ gives the unbiased consistent estimator. If r also tends to infinity the bias vanishes.

20.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a sample from $\{f_\theta : \theta \in \mathbb{R}^+\}$. Find a UMP size-$\alpha$ test of $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ for: (i) $f_\theta(x) = \theta^{-1} e^{-x/\theta} \mathbf{1}_{\mathbb{R}^+}(x)$; (ii) $f_\theta(x) = \theta^{-1} x^{\theta-1} \mathbf{1}_{(0,1)}(x)$; (iii) $f_\theta(x) = 4\theta^{-4} x^3 e^{-(x/\theta)^4} \mathbf{1}_{\mathbb{R}^+}(x)$; (iv) $f_\theta(x) = (2\pi\theta)^{-1/2} e^{-(x-1)^2/\theta} \mathbf{1}_{\mathbb{R}}(x)$.

Soluzione. First check normalization: the formula in (ii) integrates to $1/\theta^2$ rather than one. The natural normalized model is $\theta x^{\theta-1}$. The formula in (iv) integrates to $1/\sqrt{2}$; its normalized version is $(\pi\theta)^{-1/2} e^{-(x-1)^2/\theta}$, i.e. $N(1, \theta/2)$. There is no probability model with the two printed formulas literally unchanged. Under these explicitly identified normalizations the following exact UMP tests apply.

(i) $T = \sum X_i$, and $L(\theta_1)/L(\theta_0) = (\theta_0/\theta_1)^n \exp[T(1/\theta_0 - 1/\theta_1)]$ increases in T for $\theta_1 > \theta_0$. Under θ_0, $T/\theta_0 \sim \text{Gamma}(n, 1)$ (shape, rate). Reject if $T/\theta_0 > G_{n,1-\alpha}$.

(ii) Put $T = -\sum \log X_i$. Under the normalized model $-\log X_i \sim \text{Exp}(\theta)$. The likelihood ratio for larger rate decreases in T, so reject if $\theta_0 T < G_{n,\alpha}$. This lower-tail region has exact null size; larger rates make T stochastically smaller.

(iii) $W_i = X_i^4$ has survival e^{-w/θ^4}, hence is exponential with scale θ^4. The likelihood ratio increases in $T = \sum X_i^4$. Reject if $T/\theta_0^4 > G_{n,1-\alpha}$.

(iv) Under the normalized variance-$\theta/2$ model, $T = \sum (X_i - 1)^2$ has $2T/\theta_0 \sim \chi^2_n$. Larger variance gives an increasing likelihood ratio in T. Reject if $2T/\theta_0 > \chi^2_{n,1-\alpha}$. If the intended normal law was

instead $N(1,\theta)$, one must correct the exponent to $-(x-1)^2/(2\theta)$ and use $T/\theta_0 > \chi^2_{n,1-\alpha}$. In all cases stochastic order puts the maximum null size at θ_0, and NP proves the same region MP for each larger parameter, hence UMP.

Chapter 21

23 June 2021

21.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from the Uniform distribution on $(\theta, \theta + |\theta|)$. Find the MLE of θ when (i) $\theta > 0$; (ii) $\theta < 0$; (iii) $\theta \in \mathbb{R} \setminus \{0\}$.

Soluzione. For positive θ, the support is $(\theta, 2\theta)$ and likelihood θ^{-n} on the admissible interval $M/2 < \theta < m$, where $m = \min x_i$, $M = \max x_i$. It decreases with θ, so the supremum is at the lower boundary $M/2$. Using endpoint-included density versions the conventional MLE is $M/2$, provided all data are positive and $M/2 \leq m$; data violating this have zero likelihood throughout the model. For strict inequalities there is no attained MLE, only a supremum.

For negative θ, the support is $(\theta, 0)$ and likelihood $(-\theta)^{-n}$ for $\theta < m < 0$. It increases as θ approaches zero, so the supremum is as $\theta \uparrow m$; the endpoint convention gives $\hat{\theta} = m$. In the union parameter space, the observed sign identifies the branch almost surely: all positive data select $M/2$, all negative data select m. Mixed signs or zeros have zero likelihood for all parameters. Endpoint conventions do not change sampling laws but do affect literal attainment at observed extreme values.

21.2 Question 2

Enunciato. Let $((Y_1, Z_1), \dots, (Y_n, Z_n))$ be a random sample of independent exponential variables Y, Z with scale parameters $\lambda > 0, \mu > 0$. (i) Find the MLE of (λ, μ). (ii) Suppose we observe only (X_i, Δ_i), where $X_i = \min\{Y_i, Z_i\}$ and $\Delta_i = 1$ if $X_i = Y_i$, zero if $X_i = Z_i$. Find the MLE of (λ, μ).

Soluzione. For complete observations, $L = \lambda^{-n}\mu^{-n} \exp[-\sum y_i/\lambda - \sum z_i/\mu]$. Differentiating gives $\hat{\lambda} = \bar{Y}$, $\hat{\mu} = \bar{Z}$. For reduced observations, independence and continuity give joint densities

$$f(x, 1) = \lambda^{-1} e^{-x(1/\lambda + 1/\mu)}, \qquad f(x, 0) = \mu^{-1} e^{-x(1/\lambda + 1/\mu)}, \quad x > 0.$$

Set $D = \sum \Delta_i$, $T = \sum X_i$. Then $\ell = -D \log \lambda - (n - D) \log \mu - T/\lambda - T/\mu$. For $0 < D < n$ the score equations yield $\hat{\lambda} = T/D$, $\hat{\mu} = T/(n - D)$. In rate parameters $a = 1/\lambda, b = 1/\mu$, the likelihood is $a^D b^{n-D} e^{-(a+b)T}$ and is strictly concave in the positive rates, confirming the maximum. For $D = 0$, $\hat{b} = n/T$ and the supremum has $a \downarrow 0$ (infinite scale λ); for $D = n$ the roles reverse. No finite joint MLE exists at those boundary samples.

21.3 Question 3

Enunciato. Let X be a sample from a population P, and let P_0, P_1 be two known populations. Suppose T^* is a UMP test of size $\alpha \in (0, 1)$ for $H_0 : P = P_0$ versus $H_1 : P = P_1$, with power $\beta < 1$

under P_1. Show that $1 - T^*$ is a UMP test of size $1 - \beta$ for the reversed hypotheses.

Soluzione. For two simple hypotheses "UMP" means MP. By NP T^* rejects where $f_1 > cf_0$, accepts where $f_1 < cf_0$ and may randomize on equality, with $E_0T^* = \alpha$, $E_1T^* = \beta$. The complement has $E_1(1 - T^*) = 1 - \beta$ and power $E_0(1 - T^*) = 1 - \alpha$. It rejects where $f_0 > c^{-1}f_1$ and randomizes on equality, so it is an NP test for the reversed problem. If $c = 0$ handle null/alternative-zero-density sets by the same pointwise inequality, or directly note that any reversed competitor with greater P_0 power would complement to an original test of smaller than α size and at least β power; appropriate boundary randomization yields the contradiction. More rigorously NP's inequality $(T^* - T)(f_1 - cf_0) \geq 0$ rewritten for complements establishes maximal reversed power for positive finite c. Degenerate zero-density cases give the same conclusion with free rejection regions.

An explicit optimality proof avoids any ambiguity about the reversed size. Let ψ be a competitor with $E_1\psi \leq 1 - \beta$ and suppose $E_0\psi > 1 - \alpha$. Its complement $\phi = 1 - \psi$ would satisfy $E_1\phi \geq \beta$ and $E_0\phi < \alpha$. The Neyman–Pearson comparison for T^* gives

$$E_1(T^* - \phi) \geq c\, E_0(T^* - \phi) > 0,$$

contradicting $E_1\phi \geq \beta = E_1T^*$. The threshold has $c > 0$ because $c = 0$ would reject throughout the alternative support and yield $\beta = 1$; a finite threshold can be chosen for $\alpha > 0$. Hence no reversed competitor has greater power, proving the assertion for all tests, including randomized ones.

Chapter 22

9 July 2021

22.1 Question 1

Enunciato. Let $(X_1, \ldots, X_n)$ be a random sample from the discrete Uniform distribution on $\{1, \ldots, \theta\}$, $\theta \in \mathbb{N}$, and let $M = X_{(n)}$. (i) For $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$, show that $T^*(X) = 1$ if $M > \theta_0$ and $T^*(X) = \alpha$ otherwise is UMP of size α. (ii) For $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$, show that $T^*(X) = 1$ if $M > \theta_0$ or $M \leq \theta_0 \alpha^{1/n}$ and zero otherwise is UMP of size α. (iii) Show that the results still hold for a continuous Uniform distribution on $(0, \theta)$.

Soluzione. Assume integer θ_0 in the discrete model. (i) Under any null all observations are within $\{1, \ldots, \theta_0\}$ and rejection probability equals α. Against $\theta_1 > \theta_0$, observations outside that set have zero null mass and are always rejected. On the common support the likelihood ratio is the constant $(\theta_0/\theta_1)^n$, so any allocation with null expectation α is MP by NP. The proposed rule has power $1 - (1 - \alpha)(\theta_0/\theta_1)^n$ and is UMP.

(ii) As printed the discrete assertion is generally false: its null size is $[\lfloor \theta_0 \alpha^{1/n} \rfloor / \theta_0]^n$, generally strictly below α. Exact size requires randomization at a boundary. The required boundary randomization can be supplied explicitly. For example $\theta_0 = 3, n = 1, \alpha = 1/3$: against $\theta = 1$, MP requires rejection at observation 1 with probability one; against $\theta = 2$, any allocation of total null mass $1/3$ inside $\{1, 2\}$ gives the same maximum power, so that rule remains MP. For arbitrary levels the exact-size nested rule is obtained as follows: use t with $(t/\theta_0)^n \leq \alpha < ((t+1)/\theta_0)^n$, reject for $M \leq t$, randomize at $M = t + 1$ with

$$\gamma = \frac{\alpha - (t/\theta_0)^n}{((t+1)/\theta_0)^n - (t/\theta_0)^n},$$

and always reject $M > \theta_0$. For alternatives $\theta < \theta_0$, NP favors their support $M \leq \theta$, on which the ratio is constant. This nested allocation maximizes probability in every such support; for alternatives $\theta > \theta_0$, NP only fixes null size and free rejection outside the support. Hence the corrected randomized rule is UMP against both sides. The printed unrandomized formula has exact size only when $\theta_0 \alpha^{1/n}$ is an integer.

(iii) In the continuous model there is no discretization gap: $P_{\theta_0}(M \leq \theta_0 \alpha^{1/n}) = \alpha$. Below θ_0 the density ratio is constant on the smaller support and zero outside, so rejecting as far into the lower support as allowed is MP; above θ_0 the ratio is constant within the null support and infinite outside. The same lower-tail region plus automatic upper-support rejection is MP for every alternative. Its power is one when $\theta \leq c$, $(c/\theta)^n$ when $c < \theta \leq \theta_0$, and $1 - (1 - \alpha)(\theta_0/\theta)^n$ when $\theta > \theta_0$, where $c = \theta_0 \alpha^{1/n}$. The randomized rule in (i) also remains UMP in the continuous one-sided model.

22.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from a Gaussian distribution with parameter $(\mu, 1)$. Instead of recording observations, one records only whether observations are less than zero. Find the MLE of μ based on the recorded information.

Soluzione. Let $K = \sum_i \mathbf{1}_{\{X_i<0\}}$. For $X_i \sim N(\mu, 1)$ the observed indicators are iid Bernoulli with success probability $q(\mu) = \Phi(-\mu)$. Their likelihood is $q^K(1-q)^{n-K}$, with score $K/q-(n-K)/(1-q)$ in the q parametrization. For $0 < K < n$, $\hat{q} = K/n$ and

$$\hat{\mu} = -\Phi^{-1}(K/n).$$

For $K = 0$ the supremum is approached as $\mu \to +\infty$, and for $K = n$ as $\mu \to -\infty$; no finite MLE exists in those cases. For any fixed finite true μ, the strong LLN gives $K/n \to \Phi(-\mu) \in (0, 1)$, proving eventual finiteness and strong consistency. The derivative of $-\Phi^{-1}(q)$ is $-1/\phi(\Phi^{-1}(q))$, so

$$\sqrt{n}(\hat{\mu} - \mu) \Rightarrow N\left(0, \frac{q(1-q)}{\phi(\mu)^2}\right).$$

The recorded sample loses information compared with the original measurements; the information per indicator is $\phi(\mu)^2/[q(1-q)]$.

Chapter 23

9 September 2021

23.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from a regular parametric family $f_X(x;\theta)$. Describe finite and asymptotic efficiency for an estimator T_n of θ. State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator of θ.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\mathrm{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T-\theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T-S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

23.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $f_X(x;\theta) = \exp\{-(x-\theta)\}\mathbf{1}_{(\theta,+\infty)}(x)$. Find the MLE of θ, check finite-sample and asymptotic unbiasedness and consistency, and provide an optimal size-α rejection region for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$.

Soluzione. Put $M = X_{(1)}$. The likelihood is $L(\theta) = e^{-\sum x_i + n\theta}\mathbf{1}_{\{\theta < M\}}$. It increases throughout its admissible parameter set. With the literal strict support it has no attained maximum; its supremum

is approached as $\theta \uparrow M$. The conventional endpoint MLE, using an equivalent density version with $x \geq \theta$, is $\hat{\theta} = M$. Writing $X_i = \theta + E_i$, where E_i are iid Exp(1), gives $M - \theta \sim \text{Exp}(n)$. Hence

$$E\hat{\theta} = \theta + 1/n, \quad \text{Var}\,\hat{\theta} = 1/n^2, \quad \text{MSE}\,\hat{\theta} = 2/n^2.$$

It is biased at finite n, asymptotically unbiased and mean-square consistent. The estimator $M - 1/n$ is unbiased. For $H_0 : \theta = \theta_0$ versus $\theta_1 > \theta_0$, the likelihood ratio is zero if $M < \theta_1$ and constant $e^{n(\theta_1 - \theta_0)}$ on $M \geq \theta_1$. Thus rejecting in a largest-M region is NP optimal. Choose

$$c = \theta_0 - \frac{\log \alpha}{n}, \qquad C = \{M \geq c\}.$$

Under the null $P(C) = e^{-n(c-\theta_0)} = \alpha$. Under $\theta > \theta_0$ its power is $e^{-n(c-\theta)}$ if $\theta < c$, and one if $\theta \geq c$. If $c < \theta_1$, the region contains the entire alternative support and has power one, so it is still most powerful. The same region is optimal for every positive shift, hence is UMP for the one-sided composite alternative.

Chapter 24

January 2022 – handwritten recollection

24.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ be Gaussian observations with parameters $(\mu, 1)$ truncated to (α, β), where $\alpha < \beta$ are known. Show that the sample mean is the unique maximum likelihood estimator of $\theta = E[X]$. Then show that it is asymptotically efficient. The handwritten recollection adds: "or something like that, some properties when $n \to \infty$".

Soluzione. The normalized density is

$$f_\mu(x) = \frac{e^{-(x-\mu)^2/2}}{\sqrt{2\pi}[\Phi(\beta-\mu) - \Phi(\alpha-\mu)]} \mathbf{1}_{(\alpha,\beta)}(x) = h(x) e^{\mu x - a(\mu)}.$$

The cumulant $a(\mu) = \mu^2/2 + \log[\Phi(\beta-\mu) - \Phi(\alpha-\mu)]$ up to a constant has derivative $a'(\mu) = m(\mu) = EX$ and second derivative $a''(\mu) = \operatorname{Var}_\mu X = v(\mu) > 0$. Thus m is strictly increasing, with limits α, β as $\mu \to -\infty, +\infty$. The score is $n[\bar{X} - m(\mu)]$ and Hessian $-nv(\mu) < 0$. Since observations lie strictly in the truncation interval, $\bar{X} \in (\alpha, \beta)$, and there is a unique finite solution $\hat{\mu} = m^{-1}(\bar{X})$, a global maximum. By MLE invariance the estimator of $\theta = m(\mu)$ is $\hat{\theta} = \bar{X}$.

LLN gives strong consistency, CLT gives $\sqrt{n}(\bar{X} - \theta) \Rightarrow N(0, v(\mu))$. For the functional $m(\mu)$, information per observation is $v(\mu)$, so its regular asymptotic bound is $[m'(\mu)]^2/I(\mu) = v(\mu)$. In fact the mean is unbiased with variance $v(\mu)/n$ and score $U_n = n(\bar{X} - \theta)$, attaining the functional finite-sample CR bound too. The sample mean estimates the truncated mean θ, not the untruncated location μ.

24.2 Question 2

Enunciato. Let $(Y_1, \dots, Y_n) \sim U(0, \theta)$, $\theta \in (1, +\infty)$. The only information recorded is $X_i = Y_i \mathbf{1}_{[1,+\infty)}(Y_i) + \mathbf{1}_{(0,1)}(Y_i)$. Find a UMP size-$\alpha$ test of $H_0 : \theta \le \theta_0$ versus $H_1 : \theta > \theta_0$, where $\alpha \in (0, 1)$.

Soluzione. The observed variable is $X_i = \max(1, Y_i)$ almost surely: it has an atom $P(X_i = 1) = 1/\theta$ and density $1/\theta$ on $(1, \theta)$. Relative to the measure combining an atom at one with Lebesgue measure above one, the sample likelihood is $\theta^{-n} \mathbf{1}_{\{M \le \theta\}}$, $M = \max X_i \ge 1$. Assume $\theta_0 > 1$ so the null is nonempty. For any simple alternative $\theta_1 > \theta_0$, the ratio is constant $(\theta_0/\theta_1)^n$ within the null support and infinite on $M > \theta_0$. Reject if $M > \theta_0$, and otherwise reject with independent probability α. Every null parameter has rejection probability α, so the size is exact. Against $\theta > \theta_0$, $P_\theta(M \le \theta_0) = (\theta_0/\theta)^n$, giving power $1 - (1-\alpha)(\theta_0/\theta)^n$. By NP the test maximizes power for every larger alternative, hence

is UMP. If $\theta_0 \leq 1$ there is no parameter satisfying the null, so the usual size calibration is not a nonempty testing problem.

Chapter 25

15 June 2022

25.1 Question 1

Enunciato. Consider independent Bernoulli variables $(Y_1, \dots, Y_n)$ with success probabilities $w_i = e^{\alpha+\beta x_i}/(1 + e^{\alpha+\beta x_i})$, where x_i are known nonrandom constants and (α, β) are unknown. Provide maximum likelihood estimators for (α, β).

Soluzione. Writing $\eta_i = \alpha + \beta x_i$, the log likelihood is $\ell = \sum[y_i\eta_i - \log(1 + e^{\eta_i})]$. An interior MLE solves

$$\sum(y_i - w_i) = 0, \quad \sum x_i(y_i - w_i) = 0.$$

With design matrix rows $(1, x_i)$ and $W = \operatorname{diag}(w_i(1 - w_i))$, the gradient is $X^T(y - w)$ and Hessian $-X^TWX$. Full column rank makes it negative definite for finite parameters, so a finite stationary point is the unique maximum. There is generally no closed form. Newton/IRLS iterates

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix}_{new} = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}_{old} + (X^TWX)^{-1}X^T(y - w).$$

If all x_i are equal the two parameters are not separately identifiable. Complete or quasi-complete separation can drive coefficients to infinity: a finite MLE is not guaranteed. When it exists, these equations with a converged concave maximization specify it fully.

25.2 Question 2

Enunciato. Let $(X_1, \dots, X_n)$ be a random sample from $f_\theta(x) = \theta x^{-2}\mathbf{1}_{(\theta,+\infty)}(x)$. Find a UMP size-$\alpha$ test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.

Soluzione. Normalization requires $\theta > 0$. Put $M = \min X_i$. The likelihood is $\theta^n \prod x_i^{-2}\mathbf{1}_{\{M>\theta\}}$. For $\theta_1 > \theta_0$ the ratio is $(\theta_1/\theta_0)^n$ on $M > \theta_1$, zero on $\theta_0 < M \le \theta_1$. Under the null $P(M > c) = (\theta_0/c)^n$ for $c \ge \theta_0$. Rejecting $M > c$ with $c = \theta_0\alpha^{-1/n}$ has exact size α and is MP: if $c \ge \theta_1$ it uses the constant positive-ratio region, and if $c < \theta_1$ it contains the full alternative support, giving power one.

For $\theta_1 < \theta_0$ the alternative-only region $M \le \theta_0$ costs no null size and must be rejected. On $M > \theta_0$ the ratio is constant $(\theta_1/\theta_0)^n$, so reject there independently with probability α. Its alternative power is $1 - (1 - \alpha)(\theta_1/\theta_0)^n$. Any exact-size allocation on the null support is equally MP for that lower alternative. If $\theta_1 = \theta_0$, hypotheses coincide and every size-α test has power α.

Chapter 26

4 July 2022

26.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ have density or mass f_θ. Find the MLE when (i) the law is Bernoulli with $\theta \in [1/2, 3/4]$; (ii) the law is Gaussian with parameters (θ, θ^2), $\theta \in \mathbb{R} \setminus \{0\}$.

Soluzione. (i) The Bernoulli log likelihood $k \log \theta + (n-k) \log(1-\theta)$ is concave and has unconstrained maximum k/n. Thus $\hat{\theta} = \min(3/4, \max(1/2, k/n))$, including all-success/all-failure cases.

(ii) Put $A = \sum x_i$, $B = \sum x_i^2$. Apart from constants,

$$\ell(\theta) = -n \log |\theta| - \frac{B}{2\theta^2} + \frac{A}{\theta} - \frac{n}{2}, \quad \ell'(\theta) = \frac{-n\theta^2 - A\theta + B}{\theta^3}.$$

For $B > 0$ the candidates are $\theta_\pm = (-A \pm \sqrt{A^2 + 4nB})/(2n)$, one positive and one negative. The likelihood tends to zero as $\theta \to 0$ or $|\theta| \to \infty$, so each half-line has its maximum at its candidate. For any $t > 0$, $\ell(t) - \ell(-t) = 2A/t$. Therefore if $A > 0$ the global maximizer is θ_+; if $A < 0$ it is θ_-; if $A = 0$, both $\pm\sqrt{B/n}$ maximize. If all observations are zero ($B = 0$), likelihood is unbounded as $\theta \to 0$ and no MLE is attained; this event has probability zero in the stated continuous model.

26.2 Question 2

Enunciato. Let F, G be known cumulative distribution functions on $\mathbb{R}$. A single observation has distribution $\theta F + (1-\theta)G$, $\theta \in [0,1]$. Test $H_0 : \theta \in [\theta_1, \theta_2]$ versus $H_1 : \theta \notin [\theta_1, \theta_2]$, where $0 < \theta_1 \le \theta_2 < 1$. Show that a UMP test does not exist.

Soluzione. For any randomized test ϕ, put $a = E_F \phi$, $b = E_G \phi$. Its power is the affine function $\beta(\theta) = b + (a-b)\theta$. If $\theta_1 < \theta_2$ and a UMP level-α test existed, it must dominate the always-randomize test $\phi \equiv \alpha$ at both alternatives $\theta = 0$ and $\theta = 1$, so $b \ge \alpha$ and $a \ge \alpha$. The null constraint forces $a = b = \alpha$. If $F \neq G$, choose a bounded indicator h with $E_F h \neq E_G h$ and a small $t > 0$; $\phi_+ = \alpha + t[h - E_{\theta_2} h]$, after selecting the sign so the slope is positive, is in $[0,1]$, has level α at θ_2 and strictly greater power at $\theta = 1$. This contradicts the constant-power candidate. A negative-slope perturbation analogously favors $\theta = 0$. For $\theta_1 = \theta_2$ use the same perturbations centered at the single null.

However the printed assertion needs $F \neq G$. If $F = G$, distributions do not depend on θ and $\phi \equiv \alpha$ is UMP (all exact-size tests have constant power). Thus nonexistence is correct for distinguishable component distributions, not without that qualification.

Here the meaningful nonexistence assertion assumes $0 < \alpha < 1$; at trivial levels the usual tradeoff argument is inapplicable. To make the perturbation concrete, select $h = \mathbf{1}_A$ or $h = -\mathbf{1}_A$ so that

$d = E_F h - E_G h > 0$, possible because $F \neq G$. Set $m_2 = E_{\theta_2} h$ and $\phi_+(x) = \alpha + t(h(x) - m_2)$ for sufficiently small positive t. Boundedness of h and $0 < \alpha < 1$ ensure $0 \leq \phi_+ \leq 1$. Its power is

$$E_\theta \phi_+ = \alpha + td(\theta - \theta_2),$$

which is at most α throughout the null interval and strictly above α at $\theta = 1$. For a single-point null the same computation holds with $\theta_1 = \theta_2$. This establishes the contradiction without relying on an unspecified favorable direction.

Chapter 27

6 July 2023

27.1 Question 1

Enunciato. Let $(X_1, \dots, X_n)$ have density $f_X(x; \mu, \sigma) = \sigma^{-1} e^{-(x-\mu)/\sigma} / [1 + e^{-(x-\mu)/\sigma}]^2$, $\mu \in \mathbb{R}$, $\sigma > 0$. Apply one step of Newton–Raphson and Fisher scoring to find the MLE of (i) μ when σ is known; (ii) σ when μ is known.

Soluzione. Set $z_i = (x_i - \mu)/\sigma$, $t_i = \tanh(z_i/2)$, $s_i = \operatorname{sech}^2(z_i/2) = 1 - t_i^2$. Since $\log f = -\log \sigma - z - 2\log(1 + e^{-z})$, differentiation gives

$$U_\mu = \sigma^{-1} \sum t_i, \quad J_\mu = (2\sigma^2)^{-1} \sum s_i.$$

At an initial guess $\mu^{(0)}$, Newton gives $\mu^{(1)} = \mu^{(0)} + 2\sigma \sum t_i / \sum s_i$. Under the true logistic law $t = 2F(X) - 1$ is Uniform$(-1, 1)$, so $I_\mu = n/(3\sigma^2)$ and Fisher scoring gives $\mu^{(1)} = \mu^{(0)} + 3\sigma \sum t_i / n$. Evaluate each sum at the initial guess.

For scale, define $A = \sum z_i t_i$. The score and observed information are

$$U_\sigma = (A - n)/\sigma, \quad J_\sigma = \sigma^{-2} \left[2A + \frac{1}{2} \sum z_i^2 s_i - n \right].$$

Thus Newton gives $\sigma^{(1)} = \sigma^{(0)} + \sigma^{(0)}(A-n)/[2A + \frac{1}{2}\sum z_i^2 s_i - n]$, when the denominator is nonzero. The expected scale information is $I_\sigma = n(\pi^2 + 3)/(9\sigma^2)$. For completeness, $ET^2 = 1/3$, $EZ^2 = \pi^2/3$, and integration by parts using $f'(z) = -tf(z)$ gives $E(Zt) = 1$ and $E(Z^2 s) = (2\pi^2 - 12)/9$; substituting in EJ yields the constant. Hence Fisher scoring gives $\sigma^{(1)} = \sigma^{(0)} + 9\sigma^{(0)}(A - n)/[n(\pi^2 + 3)]$. Both updates require an initial positive scale and may need step shortening to remain positive. A single iteration is an update toward an MLE, not generally the final maximizing estimate.

27.2 Question 2

Enunciato. Let $\bar{X} = (X_1, \dots, X_n)$ be a sample from $N(\mu_x, \sigma^2)$ and $\bar{Y} = (Y_1, \dots, Y_m)$ an independent sample from $N(\mu_y, \sigma^2)$. Describe the rejection region for $H_0 : \mu_x - \mu_y = 0$ versus $H_1 : \mu_x - \mu_y \neq 0$.

Soluzione. Let two independent normal samples have means μ_x, μ_y and sample sizes n, m. For a common unknown variance, set $R_x = \sum (X_i - \bar{X})^2$, $R_y = \sum (Y_j - \bar{Y})^2$ and $s_p^2 = (R_x + R_y)/(n + m - 2)$. Under equal means the MLE of the common mean is $(n\bar{X} + m\bar{Y})/(n + m)$; the restricted residual sum is

$$R_0 = R_x + R_y + \frac{nm}{n + m} (\bar{X} - \bar{Y})^2.$$

Under the alternative the two fitted means are the sample means. Maximizing variance gives

$$\Lambda = \left(\frac{R_x + R_y}{R_0} \right)^{(n+m)/2} = \left(1 + \frac{T^2}{n + m - 2} \right)^{-(n+m)/2}, \quad T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{1/n + 1/m}}.$$

Under H_0, $T \sim t_{n+m-2}$, because the normalized mean difference is standard Gaussian and independent of the pooled chi-square residual. Reject $|T| > t_{n+m-2,1-\alpha/2}$.

For testing equal variances with means unrestricted, the unrestricted variance MLEs are $R_x/n, R_y/m$; the restricted estimate is $(R_x + R_y)/(n + m)$. Thus

$$\Lambda = \frac{(R_x/n)^{n/2}(R_y/m)^{m/2}}{[(R_x + R_y)/(n + m)]^{(n+m)/2}}.$$

It is largest at $R_x/n = R_y/m$ and decreases in either direction. The statistic $F = (R_x/(n - 1))/(R_y/(m - 1))$ has $F_{n-1,m-1}$ null law. The LRT rejects for $F < a$ or $F > b$, where $\Lambda(a) = \Lambda(b)$ and $P(F < a) + P(F > b) = \alpha$. The common equal-tail F test uses quantiles $F_{\alpha/2}, F_{1-\alpha/2}$ and has exact size, but is not generally the same as the equal-likelihood LRT cutoffs. Without normality the exact t/F pivots fail: with finite variances the two-sample CLT and estimated standard errors yield a large-sample normal mean test; variance tests require finite fourth moments and a fourth-moment-adjusted standard error. Gaussian likelihood procedures under misspecification require sandwich covariance, not an automatic Wilks assertion.

Chapter 28

19 December 2024 – Assignment 1

28.1 Exercise 1

Enunciato. Present the asymptotic theory of statistical hypothesis testing for both regular and non-regular parametric models, including at least one example of non-regular models.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0, \theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat\theta = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \geq 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \leq \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0, 1)$ since $P((M/\theta_0)^n \leq u) = u$. Consequently $-2 \log \Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \leq \theta_0 \alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)} \mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta - \sum X_i} \mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M - \theta) \sim \text{Exp}(1)$. Under a point null, $-2 \log \Lambda = 2n(M - \theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a - b, a + b)$ with known b, admissible centers lie between $M - b$ and $m + b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

28.2 Exercise 2

Enunciato. Let $(X_1, \ldots, X_n)$ be a random sample from a Beta distribution with parameter $(\theta, 1)$, $\theta > 0$. Provide a Gaussian limit for the moment-based and maximum likelihood estimators of θ, and discuss their large-n asymptotic relative efficiency.

Soluzione. The density is $\theta x^{\theta-1}$ on $(0, 1)$, so $EX = \theta/(\theta+1)$, $\text{Var}\, X = \theta/[(\theta+1)^2(\theta+2)]$. Matching $\bar{X}$ to its mean gives $\hat{\theta}_{MM} = \bar{X}/(1 - \bar{X})$. The derivative of $g(x) = x/(1 - x)$ at the population mean is $(\theta + 1)^2$. CLT and Delta therefore give

$$\sqrt{n}(\hat{\theta}_{MM} - \theta) \Rightarrow N\left(0, \frac{\theta(\theta + 1)^2}{\theta + 2}\right).$$

The log likelihood is $n \log \theta + (\theta - 1) \sum \log X_i$, so $\hat{\theta}_{ML} = -n/\sum \log X_i$. Put $Z_i = -\log X_i$: its survival is $P(Z > z) = P(X < e^{-z}) = e^{-\theta z}$, hence $EZ = 1/\theta$, $\text{Var}\, Z = 1/\theta^2$. Applying the Delta method to $1/\bar{Z}$ gives

$$\sqrt{n}(\hat{\theta}_{ML} - \theta) \Rightarrow N(0, \theta^2).$$

With efficiency of MM relative to ML defined as v_{ML}/v_{MM}, it is $\theta(\theta+2)/(\theta+1)^2 = 1 - 1/(\theta+1)^2 < 1$. ML is asymptotically more precise and attains information bound $I(\theta)^{-1} = \theta^2$. Both are consistent by LLN; neither finite-sample unbiasedness follows from these limits.

Chapter 29

19 December 2024 – Assignment 2

29.1 Exercise 1

Enunciato. Present the large-sample asymptotic properties of maximum likelihood estimators within regular statistical models, with focus on Cramér's theorem and its proof.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

29.2 Exercise 2

Enunciato. Let (X_1, X_2) be a random sample from Beta$(\theta, 1)$, $\theta > 0$. To test $H_0 : \theta \leq 1$ versus $H_1 : \theta > 1$, consider $C = \{(x_1, x_2) \in (0,1)^2 : x_2 \geq \sqrt{x_1}\}$ and find its power function. Find the most powerful test of size $\alpha = 3/8$ for $H_0 : \theta = 1$ versus $H_1 : \theta = 3$.

Soluzione. The joint density is $\theta^2 (x_1 x_2)^{\theta-1}$. Conditioning on $X_1 = x$,

$$\beta(\theta) = \int_0^1 [1 - x^{\theta/2}] \theta x^{\theta-1} dx = 1 - \frac{\theta}{3\theta/2} = \frac{1}{3}.$$

Its power is constant for every positive θ, hence its size against the composite null is $1/3$; it does not distinguish alternatives. For the simple hypotheses the likelihood ratio is $9(x_1 x_2)^2$, so NP rejects when $X_1 X_2 > c$. Under the null the observations are Uniform$(0, 1)$ and

$$P_1(X_1 X_2 \leq c) = \int_0^c 1 \, dx + \int_c^1 (c/x) dx = c - c \log c.$$

Thus choose the unique $c \in (0, 1)$ solving $1 - c + c \log c = 3/8$, equivalently $c(1 - \log c) = 5/8$. Under general θ, $P_\theta(X_1 X_2 \leq c) = c^\theta (1 - \theta \log c)$, so power is $1 - c^\theta (1 - \theta \log c)$, larger than $3/8$ at $\theta = 3$. Since the likelihood ratio is increasing in the product for every $\theta > 1$, the same calibrated region is UMP for the corresponding one-sided composite problem as well.

Chapter 30

19 December 2024 – Assignment 3

30.1 Exercise 1

Enunciato. Present the asymptotic theory of statistical hypothesis testing for both regular and non-regular parametric models, including at least one example of non-regular models.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0, \theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat\theta = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \geq 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \leq \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0, 1)$ since $P((M/\theta_0)^n \leq u) = u$. Consequently $-2 \log \Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \leq \theta_0 \alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)} \mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta - \sum X_i} \mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M - \theta) \sim \text{Exp}(1)$. Under a point null, $-2 \log \Lambda = 2n(M - \theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a - b, a + b)$ with known b, admissible centers lie between $M - b$ and $m + b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

30.2 Exercise 2

Enunciato. Let $(X_1, \ldots, X_n)$ be a random sample from Gamma(α, θ), with known $\alpha > 0$ and unknown $\theta > 0$. Provide a Gaussian limit for the moment-based and maximum likelihood estimators of θ, and discuss their large-n asymptotic relative efficiency.

Soluzione. Use the scale convention of the worked material: $f_\theta(x) = x^{\alpha-1} e^{-x/\theta}/[\Gamma(\alpha)\theta^\alpha]$, so $EX = \alpha\theta$ and $\text{Var}\, X = \alpha\theta^2$. Matching the first moment gives $\hat{\theta}_{MM} = \bar{X}/\alpha$. The log likelihood is $-n\alpha \log \theta - \sum X_i/\theta$ plus a parameter-free term; its derivative is $-n\alpha/\theta + \sum X_i/\theta^2$, giving $\hat{\theta}_{ML} = \bar{X}/\alpha$, the same estimator. CLT gives

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2/\alpha).$$

It is exactly unbiased with variance $\theta^2/(n\alpha)$, and both strong and mean-square consistent. Fisher information per observation is α/θ^2, so it attains the finite-sample bound. Relative efficiency is one because the estimators coincide. If θ denotes rate instead, both estimators are $\alpha/\bar{X}$ and the Delta-method limiting variance remains θ^2/α, but finite-sample unbiasedness changes.

Chapter 31

Open questions – possibili domande OLD

31.1 Question 1

Enunciato. Present the foundations of parametric statistical modeling: random sampling assumptions (iid / exchangeable / Markov), parametric family F_θ, likelihood principle, and the roles of estimation, testing, and confidence sets.

Soluzione. A statistical model is a family of probability laws $\{P_\theta : \theta \in \Theta\}$ for observable data. A parametric model has a finite-dimensional parameter; a random sample normally means iid observations with common density f_θ. Their likelihood is the joint density evaluated at the observations, considered as a function of the parameter. Independence gives $L = \prod f_\theta(x_i)$. Exchangeability means the joint law is invariant under permutations but does not itself imply independence; its likelihood must use the actual joint law. A Markov model instead factors as initial density times transition densities $f_\theta(x_1) \prod_{i \geq 2} p_\theta(x_i \mid x_{i-1})$.

Identifiability means $P_\theta = P_\eta$ only if $\theta = \eta$. Normal laws with known variance and distinct means are identifiable because their expectations differ. Observing only X^2 when $X \sim N(\theta, 1)$ loses the sign: θ and $-\theta$ produce the same distribution, so the signed parameter is not identifiable. The likelihood principle compares parameter values by the observed likelihood; multiplying it by a data-only positive factor does not change evidence or maximizers. Point estimation selects a data function to estimate a parameter or functional; confidence sets are random sets calibrated by repeated-sampling coverage; testing controls rejection probabilities under null laws. A frequentist treats the parameter as fixed and studies sampling distributions, whereas a Bayesian assigns a prior and updates it by Bayes' formula. These are distinct conditioning and calibration conventions, not alternative definitions of the same probability.

31.2 Question 2

Enunciato. Derive and discuss the likelihood and log-likelihood for a generic iid sample from $f(x; \theta)$. Explain identifiability and give criteria/examples showing when a model is (non-)identifiable.

Soluzione. A statistical model is a family of probability laws $\{P_\theta : \theta \in \Theta\}$ for observable data. A parametric model has a finite-dimensional parameter; a random sample normally means iid observations with common density f_θ. Their likelihood is the joint density evaluated at the observations, considered as a function of the parameter. Independence gives $L = \prod f_\theta(x_i)$. Exchangeability means the joint law is invariant under permutations but does not itself imply independence; its likelihood must use the actual joint law. A Markov model instead factors as initial density times transition

densities $f_\theta(x_1)\prod_{i\geq 2} p_\theta(x_i \mid x_{i-1})$.

Identifiability means $P_\theta = P_\eta$ only if $\theta = \eta$. Normal laws with known variance and distinct means are identifiable because their expectations differ. Observing only X^2 when $X \sim N(\theta, 1)$ loses the sign: θ and $-\theta$ produce the same distribution, so the signed parameter is not identifiable. The likelihood principle compares parameter values by the observed likelihood; multiplying it by a data-only positive factor does not change evidence or maximizers. Point estimation selects a data function to estimate a parameter or functional; confidence sets are random sets calibrated by repeated-sampling coverage; testing controls rejection probabilities under null laws. A frequentist treats the parameter as fixed and studies sampling distributions, whereas a Bayesian assigns a prior and updates it by Bayes' formula. These are distinct conditioning and calibration conventions, not alternative definitions of the same probability.

Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat\theta_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat\theta_n - \theta_0), \quad \sqrt{n}(\hat\theta_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat\theta_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \mathrm{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log\theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat\theta = 1/\bar X, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat\theta \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat\theta - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat\theta = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat\mu = \bar X$ and $\hat\sigma^2 = n^{-1}\sum (X_i - \bar X)^2$; the latter has expectation $(n-1)\sigma^2/n$.

31.3 Question 3

Enunciato. Define the score function and Fisher information (observed and expected). Prove $\mathbb{E}_\theta[V_n'(\theta)] = 0$ under regularity, and show $I_n(\theta) = \operatorname{Var}(V_n'(\theta)) = -\mathbb{E}(V_n''(\theta))$.

Soluzione. For an iid dominated model f_θ, the likelihood, log likelihood and score are

$$L_n(\theta; x) = \prod_i f_\theta(x_i), \quad \ell_n(\theta) = \sum_i \log f_\theta(x_i), \quad U_n(\theta) = \ell_n'(\theta).$$

Assume an open parameter space, common support, differentiability and an integrable domination permitting differentiation under the integral. Then

$$E_\theta U_1 = \int \partial_\theta f_\theta(x)dx = 0, \qquad I(\theta) = E_\theta U_1^2 = -E_\theta \partial_\theta^2 \log f_\theta(X).$$

Indeed $\partial_\theta^2 \log f = f''/f - (f'/f)^2$ and $\int f'' = 0$. Independence gives $I_n = nI$. Observed information is $J_n = -\ell_n''$, a random quantity, whereas expected information is its expectation. Information quantifies local distinguishability and the precision attainable by unbiased regular estimators. For Bernoulli observations, $U_n = (\sum X_i - n\theta)/[\theta(1-\theta)]$ and $I_n = n/[\theta(1-\theta)]$. Parameter-dependent support can invalidate differentiation and hence both identities.

31.4 Question 4

Enunciato. State the regularity conditions used in likelihood asymptotics and explain precisely where each condition is used (interchanging derivative/integral, LLN/CLT steps, Taylor expansions, etc.).

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \le 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat\theta_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat\theta_n - \theta_0), \quad \sqrt{n}(\hat\theta_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat\theta_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \mathrm{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log\theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat\theta = 1/\bar X, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat\theta \to \theta$. Since $\mathrm{Var}\, X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat\theta - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat\theta = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat\mu = \bar X$ and $\hat\sigma^2 = n^{-1}\sum (X_i - \bar X)^2$; the latter has expectation $(n-1)\sigma^2/n$.

31.5 Question 5

Enunciato. Develop maximum likelihood estimation in regular parametric models: existence and/or uniqueness issues, likelihood equations, and second-order conditions. Include at least two complete MLE derivations (e.g., Bernoulli, Poisson, Gaussian).

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\text{Var}\, X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

31.6 Question 6

Enunciato. Exponential families: define EF and natural parametrization, show how sufficient statistics arise, and derive $\mathbb{E}[A(X)]$ and $\text{Var}(A(X))$ from the EF form (including the natural parameter case).

Soluzione. Write a one-parameter exponential family as $f_\theta(x) = h(x) \exp\{\eta(\theta)A(x) - \psi(\theta)\}$ on fixed support. In the natural parameter $t = \eta(\theta)$ the log partition is $a(t) = \log \int h(x) e^{tA(x)} dx$. Differentiating under the integral gives

$$a'(t) = E_t A(X), \quad a''(t) = \text{Var}_t\, A(X) \geq 0.$$

The sample likelihood depends on data only through $T = \sum A(X_i)$, proving sufficiency by factorization. Its natural score is $T - na'(t)$, information is $na''(t)$ and the likelihood equation is $a'(\hat{t}) = T/n$. Strict positive variance makes a strictly convex and the log likelihood strictly concave, hence at most one interior maximum. A boundary maximum may still occur. Bernoulli has $A = x$, $t = \log[\theta/(1-\theta)]$, $a = \log(1 + e^t)$, giving mean θ and variance $\theta(1-\theta)$. Poisson has $A = x$, $t = \log \lambda$, $a = e^t$, giving both mean and variance λ.

31.7 Question 7

Enunciato. Present the Fisher–Neyman factorization theorem (and its corollary). Prove it (discrete or continuous version) and apply it to identify sufficient statistics in: Bernoulli, Poisson, Gaussian (unknown mean/variance cases).

Soluzione. A statistic T is sufficient for θ if the conditional law of the complete sample given T is independent of θ. In a dominated model the Fisher–Neyman factorization theorem says this is equivalent to

$$f_\theta(x) = g_\theta(T(x))h(x).$$

In the discrete case, for $T = t$ the conditional probability is $g_\theta(t)h(x)/\sum_{T(y)=t} g_\theta(t)h(y)$, so the parameter cancels. Conversely a parameter-free conditional law times the marginal law of T gives the factorization. In the dominated continuous case use conditional distributions and disintegration; the same cancellation holds almost everywhere.

For Bernoulli or Poisson sampling, $\sum X_i$ is sufficient. Gaussian sampling with known variance has sufficient $\sum X_i$ for the mean; with both mean and variance unknown $(\sum X_i, \sum X_i^2)$ is sufficient. For an exponential family $f_\theta = h \exp\{\eta(\theta)A - \psi(\theta)\}$, the sample sum of A is sufficient.

A statistic is minimal sufficient if it is a measurable function of every sufficient statistic, up to common null sets. In a dominated positive model, the likelihood-ratio criterion states that T is minimal sufficient when $T(x) = T(y)$ iff $f_\theta(x)/f_\theta(y)$ is independent of θ. Any sufficient factorization forces that parameter-free ratio within its fibers; therefore its partition refines that of T. Conversely the ratio criterion provides a factorization by choosing representatives (with standard measurability conditions). For Gaussian mean with known variance the ratio depends on θ exactly through $\sum x_i - \sum y_i$, so the sample sum is minimal sufficient. This theorem concerns minimal sufficiency and is distinct from the completeness-based UMVU theorem also associated with Lehmann and Scheffé.

Rao–Blackwell: for a square-integrable estimator U, let $V = E[U \mid T]$ with sufficient T. The conditional expectation can be chosen independently of the unknown parameter, so V is an estimator; $EV = EU$. Variance decomposition gives $\operatorname{Var} U = \operatorname{Var} V + E \operatorname{Var}(U \mid T) \geq \operatorname{Var} V$. For any target $g(\theta)$, conditioning also gives smaller squared-error risk. In Bernoulli sampling start with $U = X_1$, an unbiased estimator of θ. Given $T = \sum X_i = t$, exchangeability gives $V = t/n = \bar{X}$. Its variance is $\theta(1-\theta)/n$, strictly smaller than that of X_1 for $n > 1$ and $0 < \theta < 1$; it attains the Cramér–Rao bound. Completeness of the binomial statistic further makes it the unique UMVU estimator.

31.8 Question 8

Enunciato. Minimal sufficiency: give the definition via partitions and via the likelihood ratio criterion. State and prove the Lehmann–Scheffé minimal sufficiency theorem, and apply it to at least one model from the program.

Soluzione. A statistic T is sufficient for θ if the conditional law of the complete sample given T is independent of θ. In a dominated model the Fisher–Neyman factorization theorem says this is equivalent to

$$f_\theta(x) = g_\theta(T(x))h(x).$$

In the discrete case, for $T = t$ the conditional probability is $g_\theta(t)h(x)/\sum_{T(y)=t} g_\theta(t)h(y)$, so the parameter cancels. Conversely a parameter-free conditional law times the marginal law of T gives the factorization. In the dominated continuous case use conditional distributions and disintegration; the same cancellation holds almost everywhere.

For Bernoulli or Poisson sampling, $\sum X_i$ is sufficient. Gaussian sampling with known variance has sufficient $\sum X_i$ for the mean; with both mean and variance unknown $(\sum X_i, \sum X_i^2)$ is sufficient. For an exponential family $f_\theta = h \exp\{\eta(\theta)A - \psi(\theta)\}$, the sample sum of A is sufficient.

A statistic is minimal sufficient if it is a measurable function of every sufficient statistic, up to common null sets. In a dominated positive model, the likelihood-ratio criterion states that T is minimal sufficient when $T(x) = T(y)$ iff $f_\theta(x)/f_\theta(y)$ is independent of θ. Any sufficient factorization forces that parameter-free ratio within its fibers; therefore its partition refines that of T. Conversely the ratio criterion provides a factorization by choosing representatives (with standard measurability conditions). For Gaussian mean with known variance the ratio depends on θ exactly through $\sum x_i - \sum y_i$, so the sample sum is minimal sufficient. This theorem concerns minimal sufficiency and is distinct from the completeness-based UMVU theorem also associated with Lehmann and Scheffé.

Rao–Blackwell: for a square-integrable estimator U, let $V = E[U \mid T]$ with sufficient T. The conditional expectation can be chosen independently of the unknown parameter, so V is an estimator; $EV = EU$. Variance decomposition gives $\operatorname{Var} U = \operatorname{Var} V + E \operatorname{Var}(U \mid T) \geq \operatorname{Var} V$. For any target $g(\theta)$, conditioning also gives smaller squared-error risk. In Bernoulli sampling start with $U = X_1$, an unbiased estimator of θ. Given $T = \sum X_i = t$, exchangeability gives $V = t/n = \bar{X}$. Its variance is $\theta(1-\theta)/n$, strictly smaller than that of X_1 for $n > 1$ and $0 < \theta < 1$; it attains the Cramér–Rao bound. Completeness of the binomial statistic further makes it the unique UMVU estimator.

31.9 Question 9

Enunciato. Properties of estimators: unbiasedness, bias, variance, MSE, efficiency. Show the bias of the uncorrected sample variance and prove the unbiasedness of the corrected variance.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\operatorname{MSE}_\theta(T) = \operatorname{Var}_\theta T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

$$\operatorname{Cov}_\theta(T, U_n) = \partial_\theta E_\theta T = g'(\theta) + b'(\theta).$$

Cauchy–Schwarz therefore proves

$$\operatorname{Var}_\theta T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\operatorname{Var} T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

For iid observations with mean μ and variance σ^2, write $Q = \sum(X_i - \bar{X})^2 = \sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations gives $EQ = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. Therefore Q/n has bias $-\sigma^2/n$, while $S^2 = Q/(n-1)$ is unbiased for $n > 1$. MSE equals variance plus squared bias, so unbiasedness alone does not settle optimality.

For Gaussian observations an orthogonal change of coordinates separates $\sqrt{n}(\bar{X} - \mu)/\sigma \sim N(0,1)$ from $n-1$ independent standard normal residual coordinates. Their squared sum is $Q/\sigma^2 \sim \chi^2_{n-1}$

and is independent of the mean. A chi-square χ^2_k is the sum of squares of k independent standard normals (equivalently Gamma shape $k/2$, scale 2). If $Z \sim N(0,1)$ is independent of $W \sim \chi^2_k$, then $Z/\sqrt{W/k} \sim t_k$. Independent chi-squares U, V give $(U/a)/(V/b) \sim F_{a,b}$. Thus $\sqrt{n}(\bar{X} - \mu)/S \sim t_{n-1}$ yields exact confidence intervals and tests for the Gaussian mean.

31.10 Question 10

Enunciato. Derive the χ^2, t, and F distributions as in the course: constructions from Gaussian variables, key properties, and at least one application (e.g., distribution of sample variance in the Gaussian model).

Soluzione. For iid observations with mean μ and variance σ^2, write $Q = \sum(X_i - \bar{X})^2 = \sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations gives $EQ = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. Therefore Q/n has bias $-\sigma^2/n$, while $S^2 = Q/(n-1)$ is unbiased for $n > 1$. MSE equals variance plus squared bias, so unbiasedness alone does not settle optimality.

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31.11 Question 11

Enunciato. Cramér–Rao bound: state and prove the CR inequality for unbiased estimators in regular models. Extend to the biased/general CR bound and interpret the terms $(1 + b'(\theta))^2/I_n(\theta)$.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

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Cauchy–Schwarz therefore proves

$$\mathrm{Var}_\theta\, T \geq \frac{[g'(\theta) + b'(\theta)]^2}{I_n(\theta)}, \quad I_n = EU_n^2.$$

For unbiased estimation of θ, finite-sample efficiency means attaining $1/I_n$. Equality in Cauchy–Schwarz holds precisely when $U_n = I_n(T - \theta)$ almost surely. Conversely this equality and unbiasedness imply $\mathrm{Var}\, T = 1/I_n$, so the condition is necessary and sufficient. If T and S both attain it, then $I_n(T - S) = 0$; for positive information they are identical almost surely. Integrating the equality in θ gives $\log L_n(\theta; x) = a(\theta)T(x) + b(\theta) + c(x)$ with $a' = I_n$, $b' = -\theta I_n$, explaining its connection to exponential families. Not every exponential-family functional has an unbiased efficient estimator.

For $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v(\theta))$, asymptotic efficiency in the regular class means $v = I(\theta)^{-1}$. Asymptotic relative efficiency of T_n versus S_n is the ratio of their asymptotic variances (variance of S divided by variance of T). A Gaussian limit alone does not imply convergence of actual variances; uniform integrability of squared normalized errors is needed for that interpretation. In Bernoulli sampling $\bar{X}$ is unbiased, has variance $\theta(1-\theta)/n$ and attains the bound exactly; CLT also gives its efficient Gaussian limit.

31.12 Question 12

Enunciato. Efficient estimators: prove the characterization "T_n unbiased and efficient iff $V_n'(\theta) = I_n(\theta)(T_n - \theta)$". Discuss uniqueness of efficient unbiased estimators and relate the condition to exponential families.

Soluzione. An estimator T of $g(\theta)$ is unbiased if $E_\theta T = g(\theta)$; its bias is $b(\theta) = E_\theta T - g(\theta)$ and

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta\, T + b(\theta)^2.$$

Under common support and differentiation conditions, writing $U_n = \partial_\theta \log L_n$ gives $EU_n = 0$ and

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31.13 Question 13

Enunciato. Rao–Blackwell theorem: state and prove it; explain the role of sufficiency. Give a complete worked example where Rao–Blackwell strictly improves an unbiased estimator.

Soluzione. A statistic T is sufficient for θ if the conditional law of the complete sample given T is independent of θ. In a dominated model the Fisher–Neyman factorization theorem says this is equivalent to

$$f_\theta(x) = g_\theta(T(x))h(x).$$

In the discrete case, for $T = t$ the conditional probability is $g_\theta(t)h(x)/\sum_{T(y)=t} g_\theta(t)h(y)$, so the parameter cancels. Conversely a parameter-free conditional law times the marginal law of T gives the factorization. In the dominated continuous case use conditional distributions and disintegration; the same cancellation holds almost everywhere.

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31.14 Question 14

Enunciato. Consistency: define weak/strong consistency and show relationships (mean-square $\Rightarrow$ probability). Discuss why (and under what conditions) MLEs are strongly consistent; outline the key LLN and identifiability argument.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \operatorname{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

31.15 Question 15

Enunciato. Define $o_p(\cdot)$ and $O_p(\cdot)$. Prove the main algebra rules and explain how these notions are used to control Taylor remainders in asymptotic theory.

Soluzione. $R_n = o_p(a_n)$ means $R_n/a_n \to 0$ in probability. $R_n = O_p(a_n)$ means R_n/a_n is tight: for every $\varepsilon > 0$ there exist C, N with $P(|R_n/a_n| > C) < \varepsilon$ for $n \geq N$. A distributional limit to a finite random variable implies tightness. Union bounds show sums of two $O_p(1)$ quantities are $O_p(1)$ and sums of two $o_p(1)$ quantities are $o_p(1)$. For products, choose a tightness bound C for B_n; then $P(|A_nB_n| > \eta) \leq P(|B_n| > C) + P(|A_n| > \eta/C)$, proving $o_p(1)O_p(1) = o_p(1)$; the same argument gives $O_p(1)O_p(1) = O_p(1)$. If $B_n \to b \neq 0$, then $1/B_n \to 1/b$ and is $O_p(1)$. Taylor remainder $R_n = (\hat{\theta} - \theta)^2 O_p(1)$ is $O_p(n^{-1})$ when $\hat{\theta} - \theta = O_p(n^{-1/2})$, hence $\sqrt{n}R_n = o_p(1)$.

31.16 Question 16

Enunciato. Delta method: state and prove it from a Taylor expansion in probability. Provide one nontrivial application to an MLE via invariance (e.g., $g(\hat{\theta}_n)$).

Soluzione. Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, v(\theta))$ and g is differentiable at θ. Consistency and Taylor expansion give

$$g(\hat{\theta}_n) - g(\theta) = g'(\theta)(\hat{\theta}_n - \theta) + o_p(|\hat{\theta}_n - \theta|),$$

so $\sqrt{n}[g(\hat{\theta}_n) - g(\theta)] \Rightarrow N(0, g'(\theta)^2 v(\theta))$. If $g' = 0$, this first-order limit is degenerate and a higher-order expansion is needed. For a regular MLE $v = 1/I$. A plug-in standard error is

$$s_n = \frac{|g'(\hat{\theta}_n)|}{\sqrt{nI(\hat{\theta}_n)}}.$$

An asymptotic confidence interval is $g(\hat{\theta}_n) \pm z_{1-\alpha/2} s_n$. The Wald test of $g(\theta) = g(\theta_0)$ rejects when $|g(\hat{\theta}_n) - g(\theta_0)|/s_n > z_{1-\alpha/2}$, provided the derivative and information are nonzero at the relevant parameter.

For exponential rate θ, $I = \theta^{-2}$, $\hat{\theta} = 1/\bar{X}$ and $g(\theta) = \log\theta$. Since $g' = 1/\theta$, the asymptotic variance is one: $\sqrt{n}(\log\hat{\theta} - \log\theta) \Rightarrow N(0, 1)$. The interval for log rate is $\log\hat{\theta} \pm z/\sqrt{n}$, and exponentiating yields the positive interval $[\hat{\theta}e^{-z/\sqrt{n}}, \hat{\theta}e^{z/\sqrt{n}}]$. MLE invariance follows by defining the profile likelihood of $\eta = g(\theta)$ as $\sup_{g(\theta)=\eta} L(\theta)$: the image of an MLE maximizes it, even for noninjective maps. Invariance concerns maximization, not equality of variances under reparametrization.

31.17 Question 17

Enunciato. Present and prove Cramér's theorem (asymptotic normality of the MLE) in the 1D regular case: Taylor expansion of the score, CLT for the score, LLN for the observed information, and control of the remainder term.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

31.18 Question 18

Enunciato. Hypothesis testing setup: define hypotheses (simple/composite), rejection regions, type I/II errors, size/level, power function. Discuss the trade-off between α and β with an explicit Binomial example.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0, 1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > c f_0\}} + \gamma(x) \mathbf{1}_{\{f_1 = c f_0\}}, \qquad E_0 \phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - c f_0) \geq 0$. Integrating proves

$$E_1 \phi^* - E_1 \phi \geq c(E_0 \phi^* - E_0 \phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x) \exp\{\eta(\theta) T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \text{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \leq \alpha \leq P_{\theta_0}(T \geq k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t} \theta^t (1-\theta)^{n-t} + \gamma \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

31.19 Question 19

Enunciato. Neyman–Pearson lemma: state and prove it (simple vs simple). Show how NP yields tests based on sufficient statistics and explain the monotone likelihood ratio (MLR) property and its implication for UMP tests.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0,1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

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$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t}\theta^t(1-\theta)^{n-t} + \gamma\binom{n}{k}\theta^k(1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

31.20 Question 20

Enunciato. Likelihood Ratio Test (LRT): define λ, derive LRT rejection regions, and work out at least one classical LRT completely (e.g., Gaussian mean test with unknown variance leading to a t-test).

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0,1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0 \log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

For iid $N(\mu, \sigma^2)$ with both parameters unknown, the unrestricted MLEs are $\bar X$ and R/n, $R = \sum(X_i - \bar X)^2$. Under $\mu = \mu_0$, the variance MLE is $[R + n(\bar X - \mu_0)^2]/n$. Maximized Gaussian likelihood therefore gives

$$\Lambda = \left(1 + \frac{n(\bar X - \mu_0)^2}{R}\right)^{-n/2} = \left(1 + \frac{T^2}{n-1}\right)^{-n/2}, \quad T = \frac{\sqrt{n}(\bar X - \mu_0)}{S},\ S^2 = R/(n-1).$$

Normal orthogonal projections give $\sqrt{n}(\bar X - \mu_0)/\sigma \sim N(0,1)$ independently of $R/\sigma^2 \sim \chi^2_{n-1}$, so $T \sim t_{n-1}$. The exact level-α region is $|T| > t_{n-1,1-\alpha/2}$.

31.21 Question 21

Enunciato. Present Wilks' theorem in regular parametric models: statement (dimension 1 and general dimension), precise assumptions, and a full demonstration using Taylor expansion around the MLE and the asymptotic normality of the MLE.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2 \log \Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0,1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta \phi$, size $\sup_{\Theta_0} \beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X \log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0 \log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

31.22 Question 22

Enunciato. Apply Wilks to a specific regular model (e.g., Poisson $H_0 : \lambda = \lambda_0$ vs $H_1 : \lambda \neq \lambda_0$): compute $-2\log\lambda$ explicitly and show its asymptotic χ^2 limit.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

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$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X\log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

31.23 Question 23

Enunciato. Non-regular models: define what "non-regular" means (e.g., parameter-dependent support, boundary points, non-differentiability). Explain which steps of Cramér/Wilks fail and what phenomena replace them.

Soluzione. Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0, \theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat\theta = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0, 1)$ since $P((M/\theta_0)^n \le u) = u$.

Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \text{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b, a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

31.24 Question 24

Enunciato. Develop the asymptotic theory of hypothesis testing in at least one non-regular model, including a full worked example. Suggested example: Uniform with unknown endpoints (e.g., $X_i \sim U(a-b, a+b)$ or $U(0,\theta)$), showing how the MLE depends on order statistics and deriving the (non-Gaussian / non-χ^2) asymptotic behavior of the LRT (or an alternative test).

Soluzione. Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0,\theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M\le\theta\}}, \quad M = \max X_i, \quad \hat\theta = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \le u) = u$. Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \text{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b, a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

31.25 Question 25

Enunciato. Compare regular vs non-regular asymptotics for LRTs: distributional limits of $-2\log\lambda$, scaling rates (e.g., $\sqrt{n}$ vs n), and practical consequences for critical values and p-values.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0,1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

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The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

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Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

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$$P_\theta(n(1-M/\theta) > t) = (1-t/n)^n \to e^{-t}.$$

Thus $n(\theta-M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \le u) = u$. Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

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31.26 Question 26

Enunciato. Invariance of the MLE: state and prove the invariance principle. Provide two applications from the notes (e.g., survival/binomial reparametrization; misclassification model leading to $\widehat{\theta}_n = (\widehat{q}_n - (1-p))/(2p-1)$) and discuss consistency/asymptotic normality under the transformation.

Soluzione. Suppose $\sqrt{n}(\hat\theta_n-\theta) \Rightarrow N(0,v(\theta))$ and g is differentiable at θ. Consistency and Taylor expansion give

$$g(\hat\theta_n) - g(\theta) = g'(\theta)(\hat\theta_n-\theta) + o_p(|\hat\theta_n-\theta|),$$

so $\sqrt{n}[g(\hat{\theta}_n)-g(\theta)] \Rightarrow N(0, g'(\theta)^2 v(\theta))$. If $g' = 0$, this first-order limit is degenerate and a higher-order expansion is needed. For a regular MLE $v = 1/I$. A plug-in standard error is

$$s_n = \frac{|g'(\hat{\theta}_n)|}{\sqrt{nI(\hat{\theta}_n)}}.$$

An asymptotic confidence interval is $g(\hat{\theta}_n) \pm z_{1-\alpha/2} s_n$. The Wald test of $g(\theta) = g(\theta_0)$ rejects when $|g(\hat{\theta}_n) - g(\theta_0)|/s_n > z_{1-\alpha/2}$, provided the derivative and information are nonzero at the relevant parameter.

For exponential rate θ, $I = \theta^{-2}$, $\hat{\theta} = 1/\bar{X}$ and $g(\theta) = \log\theta$. Since $g' = 1/\theta$, the asymptotic variance is one: $\sqrt{n}(\log\hat{\theta} - \log\theta) \Rightarrow N(0, 1)$. The interval for log rate is $\log\hat{\theta} \pm z/\sqrt{n}$, and exponentiating yields the positive interval $[\hat{\theta}e^{-z/\sqrt{n}}, \hat{\theta}e^{z/\sqrt{n}}]$. MLE invariance follows by defining the profile likelihood of $\eta = g(\theta)$ as $\sup_{g(\theta)=\eta} L(\theta)$: the image of an MLE maximizes it, even for noninjective maps. Invariance concerns maximization, not equality of variances under reparametrization.

Two complete examples illustrate invariance. For binomial survival data $K \sim \mathrm{Bin}(n, q)$ with survival probability $q = e^{-\theta t}$, $t > 0$, first maximize $q^K(1-q)^{n-K}$ to get $\hat{q} = K/n$. Inverting yields $\hat{\theta} = -\log(K/n)/t$ for $0 < K \le n$, with the supremum at $\theta \to \infty$ when $K = 0$. At an interior positive true rate, Delta gives variance $(1-q)/(t^2 q)$ for $\sqrt{n}(\hat{\theta} - \theta)$.

For randomized responses with a known coin probability $p \neq 1/2$, the observed success probability is $q = 1 - p + (2p-1)\theta$. The unconstrained transformed estimator is $[K/n - (1-p)]/(2p-1)$. Since $\theta \in [0, 1]$, the admissible q interval is between p and $1-p$; the actual MLE first projects K/n to that interval, then inverts. Its interior asymptotic variance is $q(1-q)/(2p-1)^2$, by the Delta method. At $p = 1/2$ the parameter is unidentifiable. At boundaries the projection changes the limit law. These examples also show why MLE invariance must respect parameter constraints.

31.27 Question 27

Enunciato. Bayesian vs frequentist learning of θ: present the Beta–Bernoulli conjugate model, derive the posterior, and derive the Bayes estimator under squared loss. Compare its large-n behavior with the MLE and moment estimator.

Soluzione. For Bernoulli observations with k successes and a Beta(a, b) prior, Bayes' formula multiplies $\theta^k(1-\theta)^{n-k}$ by $\theta^{a-1}(1-\theta)^{b-1}/B(a, b)$. Normalization gives a Beta$(a + k, b + n - k)$ posterior. Squared posterior loss is minimized by its mean: differentiating $E[(t-\theta)^2 \mid X]$ gives $t = E[\theta \mid X] = (a + k)/(a + b + n)$. The MLE and moment estimator are k/n. For fixed a, b, the difference between posterior mean and MLE is $O(1/n)$, so they share the interior $\sqrt{n}$ Gaussian limit with variance $\theta(1-\theta)$. The posterior variance is $(a + k)(b + n - k)/[(a + b + n)^2(a + b + n + 1)]$. Bayesian statements average over a prior/posterior; frequentist unbiasedness and coverage concern repeated sampling at fixed θ.

Chapter 32

Open questions – domande scritto

32.1 Question 1

Enunciato. (MLE: consistenza + asintotica normalità + efficienza) Present the asymptotic theory of the maximum likelihood estimator in a *regular* parametric model: state precise regularity conditions, prove (sketch with all key steps) *strong/weak consistency* and *Cramér theorem*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\Big(0, \frac{1}{I(\theta_0)}\Big),$$

and discuss asymptotic efficiency and the role of Fisher information. Include a complete worked example (e.g. Bernoulli/Poisson/Exponential) with explicit score, information, and asymptotic variance.

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \text{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

32.2 Question 2

Enunciato. (Wilks + LRT asintotico in modelli regolari) Develop the likelihood ratio testing framework for *regular* parametric models: define the LRT statistic λ, state Wilks theorem (1D and general dimension), and provide a full proof (or a detailed proof skeleton) showing that under H_0,

$$-2 \log \lambda(X_1, \ldots, X_n) \xrightarrow{d} \chi^2_r,$$

where r is the number of restrictions. Work out an explicit example (e.g. Poisson $H_0 : \lambda = \lambda_0$ or Gaussian mean/variance) computing $-2 \log \lambda$ and explaining how critical values are obtained.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}, \qquad D = -2 \log \Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta \phi$, size $\sup_{\Theta_0} \beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat{\theta}$ gives

$$D = -\ell_n''(\tilde{\theta})(\hat{\theta} - \theta_0)^2 = \left[-\ell_n''(\tilde{\theta})/n\right] \left[\sqrt{n}(\hat{\theta} - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat{\lambda} = \bar{X}$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar{X} \log(\bar{X}/\lambda_0) - (\bar{X} - \lambda_0)\},$$

with $0 \log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar{X} - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar{X} - \lambda_0)^2/\bar{X}$ (when $\bar{X} > 0$), score uses $n(\bar{X} - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

32.3 Question 3

Enunciato. (NP lemma + UMP in exponential families) Present the Neyman–Pearson lemma for simple vs. simple hypotheses and prove it. Then develop the construction of UMP level-α tests in one-parameter exponential families using the monotone likelihood ratio (MLR) property (and, if appropriate, sufficiency). Include at least one complete example where you: (i) identify the relevant statistic (typically $\sum A(X_i)$), (ii) construct the rejection region, (iii) calibrate the level α, (iv) compute and discuss the power function.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0,1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x)\exp\{\eta(\theta)T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \mathrm{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \leq \alpha \leq P_{\theta_0}(T \geq k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t}\theta^t(1-\theta)^{n-t} + \gamma\binom{n}{k}\theta^k(1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

32.4 Question 4

Enunciato. (Non-regular models: failure of standard asymptotics) Define what makes a parametric model *non-regular* (parameter-dependent support, boundary points, lack of differentiability, etc.). Explain precisely which steps in the proofs of Cramér theorem and Wilks theorem fail, and what changes in the asymptotic behavior (rates and limit laws). Provide at least one fully developed non-regular example (e.g. $X_i \sim U(0,\theta)$ or $X_i \sim U(a-b, a+b)$): derive the MLE (via order statistics), study its convergence rate, and analyze the asymptotic distribution of an LRT (or an alternative test), highlighting why the limit is not the usual χ^2.

Soluzione. Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0,\theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M \leq \theta\}}, \quad M = \max X_i, \quad \hat{\theta} = M, \quad P_\theta(M \leq x) = (x/\theta)^n \quad (0 \leq x \leq \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \geq 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \mathrm{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \leq \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \leq u) = u$.

Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \mathrm{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b, a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

32.5 Question 5

Enunciato. (Asymptotic tests beyond exact pivots: Wald, Score/LM, LR) Present the three classical large-sample tests in regular models: Wald, Score (Lagrange multiplier), and Likelihood Ratio. Derive each statistic from first principles (Taylor expansion of the score / observed information), state their asymptotic null distributions, and discuss their first-order asymptotic equivalence. Apply all three tests to the same model (e.g. Poisson or Bernoulli) for testing $H_0 : \theta = \theta_0$, and compare their finite-sample behavior qualitatively (e.g. dependence on parametrization, boundary issues, small n).

Soluzione. For a scalar regular model, let $U_n = \ell_n'$ and $I_n = nI$. Under $H_0 : \theta = \theta_0$, the standardized score $U_n(\theta_0)/\sqrt{nI(\theta_0)}$ tends to $N(0,1)$ by CLT. Its square is the score/LM statistic. Cramér's theorem standardizes the MLE, giving the Wald statistic $W = nI(\hat\theta)(\hat\theta-\theta_0)^2$. Expanding the score equation yields $\sqrt{n}(\hat\theta-\theta_0) = I(\theta_0)^{-1}U_n(\theta_0)/\sqrt{n} + o_p(1)$; substitution in W and the second-order likelihood expansion shows W, LM and $-2\log\Lambda$ differ by $o_p(1)$ under the null and contiguous local alternatives. Each has null limit χ^2_1. Their size is asymptotic, their small-sample rejection sets can differ, and the approximations need an interior parameter, nondegenerate information and the likelihood regularity assumptions.

For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0,1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta-\theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta-\theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \mathrm{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X\log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X-\lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X-\lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X-\lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

32.6 Question 6

Enunciato. (Delta method + invariance: inference on transformations) Assume $\widehat{\theta}_n$ is a regular MLE with Cramér asymptotics. State and prove the Delta method and use it to derive: (i) the asymptotic distribution of $g(\widehat{\theta}_n)$, (ii) an asymptotic confidence interval for $g(\theta)$, (iii) an asymptotic level-α test for $H_0 : g(\theta) = g(\theta_0)$. Include a worked example relevant for quantitative finance (e.g. log-intensity, volatility/variance transformation, or survival/hazard parametrization), and comment on how invariance interacts with asymptotic variance.

Soluzione. Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, v(\theta))$ and g is differentiable at θ. Consistency and Taylor expansion give

$$g(\hat{\theta}_n) - g(\theta) = g'(\theta)(\hat{\theta}_n - \theta) + o_p(|\hat{\theta}_n - \theta|),$$

so $\sqrt{n}[g(\hat{\theta}_n) - g(\theta)] \Rightarrow N(0, g'(\theta)^2 v(\theta))$. If $g' = 0$, this first-order limit is degenerate and a higher-order expansion is needed. For a regular MLE $v = 1/I$. A plug-in standard error is

$$s_n = \frac{|g'(\hat{\theta}_n)|}{\sqrt{nI(\hat{\theta}_n)}}.$$

An asymptotic confidence interval is $g(\hat{\theta}_n) \pm z_{1-\alpha/2} s_n$. The Wald test of $g(\theta) = g(\theta_0)$ rejects when $|g(\hat{\theta}_n) - g(\theta_0)|/s_n > z_{1-\alpha/2}$, provided the derivative and information are nonzero at the relevant parameter.

For exponential rate θ, $I = \theta^{-2}$, $\hat{\theta} = 1/\bar{X}$ and $g(\theta) = \log \theta$. Since $g' = 1/\theta$, the asymptotic variance is one: $\sqrt{n}(\log \hat{\theta} - \log \theta) \Rightarrow N(0, 1)$. The interval for log rate is $\log \hat{\theta} \pm z/\sqrt{n}$, and exponentiating yields the positive interval $[\hat{\theta} e^{-z/\sqrt{n}}, \hat{\theta} e^{z/\sqrt{n}}]$. MLE invariance follows by defining the profile likelihood of $\eta = g(\theta)$ as $\sup_{g(\theta)=\eta} L(\theta)$: the image of an MLE maximizes it, even for noninjective maps. Invariance concerns maximization, not equality of variances under reparametrization.

32.7 Question 7

Enunciato. (Sufficiency + Rao–Blackwell + efficiency in exponential families) Present the theory linking sufficiency and optimal estimation: state and prove Fisher–Neyman factorization, Rao–Blackwell theorem, and discuss how exponential family structure leads to (asymptotically) efficient estimators. Give a complete example in an exponential family where you: (i) identify a sufficient statistic, (ii) start from a naive unbiased estimator, (iii) Rao–Blackwellize it, (iv) compare variances and discuss whether the resulting estimator achieves the Cramér–Rao bound (finite-sample or asymptotically).

Soluzione. A statistic T is sufficient for θ if the conditional law of the complete sample given T is independent of θ. In a dominated model the Fisher–Neyman factorization theorem says this is equivalent to

$$f_\theta(x) = g_\theta(T(x))h(x).$$

In the discrete case, for $T = t$ the conditional probability is $g_\theta(t)h(x)/\sum_{T(y)=t} g_\theta(t)h(y)$, so the parameter cancels. Conversely a parameter-free conditional law times the marginal law of T gives the factorization. In the dominated continuous case use conditional distributions and disintegration; the same cancellation holds almost everywhere.

For Bernoulli or Poisson sampling, $\sum X_i$ is sufficient. Gaussian sampling with known variance has sufficient $\sum X_i$ for the mean; with both mean and variance unknown $(\sum X_i, \sum X_i^2)$ is sufficient. For an exponential family $f_\theta = h \exp\{\eta(\theta)A - \psi(\theta)\}$, the sample sum of A is sufficient.

A statistic is minimal sufficient if it is a measurable function of every sufficient statistic, up to common null sets. In a dominated positive model, the likelihood-ratio criterion states that T is minimal

sufficient when $T(x) = T(y)$ iff $f_\theta(x)/f_\theta(y)$ is independent of θ. Any sufficient factorization forces that parameter-free ratio within its fibers; therefore its partition refines that of T. Conversely the ratio criterion provides a factorization by choosing representatives (with standard measurability conditions). For Gaussian mean with known variance the ratio depends on θ exactly through $\sum x_i - \sum y_i$, so the sample sum is minimal sufficient. This theorem concerns minimal sufficiency and is distinct from the completeness-based UMVU theorem also associated with Lehmann and Scheffé.

Rao–Blackwell: for a square-integrable estimator U, let $V = E[U \mid T]$ with sufficient T. The conditional expectation can be chosen independently of the unknown parameter, so V is an estimator; $EV = EU$. Variance decomposition gives $\operatorname{Var} U = \operatorname{Var} V + E \operatorname{Var}(U \mid T) \geq \operatorname{Var} V$. For any target $g(\theta)$, conditioning also gives smaller squared-error risk. In Bernoulli sampling start with $U = X_1$, an unbiased estimator of θ. Given $T = \sum X_i = t$, exchangeability gives $V = t/n = \bar{X}$. Its variance is $\theta(1-\theta)/n$, strictly smaller than that of X_1 for $n > 1$ and $0 < \theta < 1$; it attains the Cramér–Rao bound. Completeness of the binomial statistic further makes it the unique UMVU estimator.

32.8 Question 8

Enunciato. (Two-sample Gaussian inference: exact + asymptotic viewpoints) Consider two independent Gaussian samples with possibly unknown parameters. Develop the likelihood approach to: (i) test equality of means under equal variances (derive the LRT and show it reduces to a t-test), (ii) test equality of variances (derive the LRT and show it reduces to an F-test). Then discuss what breaks (and what can still be done) if normality fails, motivating asymptotic alternatives.

Soluzione. Let two independent normal samples have means μ_x, μ_y and sample sizes n, m. For a common unknown variance, set $R_x = \sum(X_i - \bar{X})^2$, $R_y = \sum(Y_j - \bar{Y})^2$ and $s_p^2 = (R_x + R_y)/(n+m-2)$. Under equal means the MLE of the common mean is $(n\bar{X} + m\bar{Y})/(n+m)$; the restricted residual sum is

$$R_0 = R_x + R_y + \frac{nm}{n+m}(\bar{X} - \bar{Y})^2.$$

Under the alternative the two fitted means are the sample means. Maximizing variance gives

$$\Lambda = \left(\frac{R_x + R_y}{R_0}\right)^{(n+m)/2} = \left(1 + \frac{T^2}{n+m-2}\right)^{-(n+m)/2}, \quad T = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{1/n + 1/m}}.$$

Under H_0, $T \sim t_{n+m-2}$, because the normalized mean difference is standard Gaussian and independent of the pooled chi-square residual. Reject $|T| > t_{n+m-2,1-\alpha/2}$.

For testing equal variances with means unrestricted, the unrestricted variance MLEs are $R_x/n, R_y/m$; the restricted estimate is $(R_x + R_y)/(n+m)$. Thus

$$\Lambda = \frac{(R_x/n)^{n/2}(R_y/m)^{m/2}}{[(R_x + R_y)/(n+m)]^{(n+m)/2}}.$$

It is largest at $R_x/n = R_y/m$ and decreases in either direction. The statistic $F = (R_x/(n-1))/(R_y/(m-1))$ has $F_{n-1,m-1}$ null law. The LRT rejects for $F < a$ or $F > b$, where $\Lambda(a) = \Lambda(b)$ and $P(F < a) + P(F > b) = \alpha$. The common equal-tail F test uses quantiles $F_{\alpha/2}, F_{1-\alpha/2}$ and has exact size, but is not generally the same as the equal-likelihood LRT cutoffs. Without normality the exact t/F pivots fail: with finite variances the two-sample CLT and estimated standard errors yield a large-sample normal mean test; variance tests require finite fourth moments and a fourth-moment-adjusted standard error. Gaussian likelihood procedures under misspecification require sandwich covariance, not an automatic Wilks assertion.

Chapter 33

Open questions – full metal gasing

33.1 Question 1

Enunciato. Present the asymptotic theory of maximum likelihood estimation in regular parametric models: state regularity conditions, sketch the proof of Cramér theorem (asymptotic normality of the MLE), and discuss asymptotic efficiency (link with Fisher information).

Soluzione. Let X_i be iid from f_θ, with true interior parameter θ_0. The MLE maximizes $\ell_n(\theta) = \sum \log f_\theta(X_i)$. Sufficient consistency assumptions are identifiability, compact parameter space (or a separate tightness/coercivity argument), continuity in θ and a uniform law of large numbers for $n^{-1}\ell_n$. The population criterion satisfies

$$E_{\theta_0} \log f_\theta - E_{\theta_0} \log f_{\theta_0} = -D_{KL}(f_{\theta_0} \| f_\theta) \leq 0,$$

with equality only at θ_0. Outside any neighborhood of θ_0 its supremum is strictly smaller on a compact set. Uniform convergence transfers this separation to the empirical criterion, proving consistency of any approximate maximizer. An almost sure uniform LLN gives strong consistency.

For asymptotic normality also require common support, differentiability of the log density in a neighborhood, $0 < I(\theta_0) < \infty$, CLT for the score, LLN for second derivatives, an integrable local envelope controlling third derivatives, and an interior consistent MLE solving the score equation. Taylor's formula at a point θ_n^* between $\hat{\theta}_n$ and θ_0 gives

$$0 = U_n(\theta_0) + \ell_n''(\theta_n^*)(\hat{\theta}_n - \theta_0), \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{U_n(\theta_0)/\sqrt{n}}{-\ell_n''(\theta_n^*)/n}.$$

The numerator converges to $N(0, I(\theta_0))$ because score summands are centered with variance I. The denominator converges in probability to $I(\theta_0)$; consistency and the derivative envelope allow replacement of θ_n^* by θ_0. Slutsky yields $N(0, I^{-1})$. In the third-order formulation the remainder is bounded by a local sample average envelope times $|\hat{\theta}_n - \theta_0|$, hence is $o_p(1)$. A bounded sample average is $O_p(1)$, not necessarily $o_p(1)$. This is Cramér's theorem. It establishes first-order efficiency in the regular estimation class, not automatic finite-sample unbiasedness.

Example: for $X_i \sim \mathrm{Exp}(\theta)$ with rate $\theta > 0$,

$$\ell_n = n \log \theta - \theta \sum X_i, \quad U_n = n/\theta - \sum X_i, \quad \hat{\theta} = 1/\bar{X}, \quad I(\theta) = \theta^{-2}.$$

The strong LLN gives $\hat{\theta} \to \theta$. Since $\operatorname{Var} X = \theta^{-2}$, CLT plus the derivative of $1/x$ gives $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N(0, \theta^2)$. For $n > 1$, $E\hat{\theta} = n\theta/(n-1)$, so asymptotic efficiency coexists with finite-sample bias. For Gaussian $N(\mu, \sigma^2)$ observations, differentiating the log likelihood gives $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = n^{-1} \sum (X_i - \bar{X})^2$; the latter has expectation $(n-1)\sigma^2/n$.

33.2 Question 2

Enunciato. Present the likelihood ratio test (LRT) for composite hypotheses in regular parametric models: define the statistic λ, derive (and sketch a proof of) Wilks theorem, and explain how this yields asymptotic χ^2 critical values.

Soluzione. For hypotheses $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \notin \Theta_0$, define

$$\Lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}, \qquad D = -2\log\Lambda.$$

Small Λ, or large D, contradicts the restrictions imposed by the null. A test $\phi(x) \in [0, 1]$ rejects with that conditional probability. Its power is $\beta(\theta) = E_\theta\phi$, size $\sup_{\Theta_0}\beta$, and level α means size at most α. Type I error is rejection under the null and type II error is acceptance under an alternative.

In an interior regular d-dimensional model with a smooth null manifold of codimension r, nonsingular information and consistent restricted/unrestricted MLEs, Wilks' theorem states $D \Rightarrow \chi^2_r$ under the null. In one dimension with a fixed point null, Taylor expansion at $\hat\theta$ gives

$$D = -\ell_n''(\tilde\theta)(\hat\theta - \theta_0)^2 = \left[-\ell_n''(\tilde\theta)/n\right]\left[\sqrt{n}(\hat\theta - \theta_0)\right]^2 \Rightarrow \chi^2_1.$$

The score at the MLE vanishes, observed information divided by n tends to I, and the squared normal limit is $I^{-1}\chi^2_1$. In several dimensions the local quadratic likelihood is the squared information-metric distance to the tangent space of the null. Projection of a standard Gaussian onto the r-dimensional orthogonal complement gives χ^2_r. Reject when $D > \chi^2_{r,1-\alpha}$; the calibration is asymptotic, not exact.

For $X_i \sim \text{Pois}(\lambda)$, $\hat\lambda = \bar X$, $I(\lambda) = 1/\lambda$ and

$$D = 2n\{\bar X\log(\bar X/\lambda_0) - (\bar X - \lambda_0)\},$$

with $0\log 0 = 0$. Taylor expansion at λ_0 yields $D = n(\bar X - \lambda_0)^2/\lambda_0 + o_p(1)$; CLT proves the χ^2_1 limit. Wald uses $n(\bar X - \lambda_0)^2/\bar X$ (when $\bar X > 0$), score uses $n(\bar X - \lambda_0)^2/\lambda_0$, and LR uses the expression above. They agree to first order but differ at small samples. Wald is parametrization dependent and problematic at boundaries.

33.3 Question 3

Enunciato. Discuss UMP testing in (natural) exponential families: state the Neyman–Pearson lemma, explain how sufficiency and the monotone likelihood ratio (MLR) property lead to UMP tests for one-sided hypotheses, and give one concrete example (e.g. Bernoulli or Poisson).

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \geq 0$ and $\gamma(x) \in [0, 1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > cf_0\}} + \gamma(x)\mathbf{1}_{\{f_1 = cf_0\}}, \qquad E_0\phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - cf_0) \geq 0$. Integrating proves

$$E_1\phi^* - E_1\phi \geq c(E_0\phi^* - E_0\phi) \geq 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x)\exp\{\eta(\theta)T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \text{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \le \alpha \le P_{\theta_0}(T \ge k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t} \theta^t (1-\theta)^{n-t} + \gamma \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

33.4 Question 4

Enunciato. Present the main concepts of asymptotic testing beyond exact finite-sample distributions: define size/level and power function, explain the trade-off between type I and type II errors, and show how asymptotic approximations replace exact $t/F/\chi^2$ distributions when assumptions fail.

Soluzione. Let f_0, f_1 be densities under two simple hypotheses relative to a common dominating measure. Choose $c \ge 0$ and $\gamma(x) \in [0, 1]$ so that

$$\phi^*(x) = \mathbf{1}_{\{f_1 > c f_0\}} + \gamma(x) \mathbf{1}_{\{f_1 = c f_0\}}, \qquad E_0 \phi^* = \alpha.$$

For any level-α test ϕ, pointwise $(\phi^* - \phi)(f_1 - c f_0) \ge 0$. Integrating proves

$$E_1 \phi^* - E_1 \phi \ge c(E_0 \phi^* - E_0 \phi) \ge 0.$$

Thus ϕ^* is most powerful. Points with $f_0 = 0 < f_1$ are rejected freely. Randomization can be essential for discrete data.

If $L_n(\theta; x) = h(x) \exp\{\eta(\theta) T(x) - A_n(\theta)\}$ with increasing η, the likelihood ratio for $\theta_1 > \theta_0$ is increasing in T. A common upper-tail rejection rule is therefore MP for every simple alternative, hence UMP for $H_0 : \theta \le \theta_0$ against $H_1 : \theta > \theta_0$. Stochastic monotonicity makes the largest null rejection probability occur at θ_0. Sufficiency explains why the ratio depends only on T.

For Bernoulli data, $T = \sum X_i \sim \text{Bin}(n, \theta)$ and $\eta = \log[\theta/(1-\theta)]$. Choose integer k with $P_{\theta_0}(T > k) \le \alpha \le P_{\theta_0}(T \ge k)$ and

$$\gamma = \frac{\alpha - P_{\theta_0}(T > k)}{P_{\theta_0}(T = k)}.$$

Reject for $T > k$, reject with probability γ at $T = k$. The power is

$$\beta(\theta) = \sum_{t=k+1}^{n} \binom{n}{t} \theta^t (1-\theta)^{n-t} + \gamma \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

It is increasing; type II error is $1 - \beta(\theta)$. Increasing α enlarges rejection and raises power at the cost of more false rejections. For two-sided alternatives opposing tails generally require different MP regions, so one-sided UMP results do not transfer automatically.

For a scalar regular model, let $U_n = \ell_n'$ and $I_n = nI$. Under $H_0 : \theta = \theta_0$, the standardized score $U_n(\theta_0)/\sqrt{nI(\theta_0)}$ tends to $N(0, 1)$ by CLT. Its square is the score/LM statistic. Cramér's theorem standardizes the MLE, giving the Wald statistic $W = nI(\hat{\theta})(\hat{\theta} - \theta_0)^2$. Expanding the score equation yields $\sqrt{n}(\hat{\theta} - \theta_0) = I(\theta_0)^{-1} U_n(\theta_0)/\sqrt{n} + o_p(1)$; substitution in W and the second-order likelihood expansion shows W, LM and $-2 \log \Lambda$ differ by $o_p(1)$ under the null and contiguous local alternatives. Each has null limit χ^2_1. Their size is asymptotic, their small-sample rejection sets can differ, and the

approximations need an interior parameter, nondegenerate information and the likelihood regularity assumptions.

Let two independent normal samples have means μ_x, μ_y and sample sizes n, m. For a common unknown variance, set $R_x = \sum(X_i - \bar{X})^2$, $R_y = \sum(Y_j - \bar{Y})^2$ and $s_p^2 = (R_x + R_y)/(n+m-2)$. Under equal means the MLE of the common mean is $(n\bar{X} + m\bar{Y})/(n+m)$; the restricted residual sum is

$$R_0 = R_x + R_y + \frac{nm}{n+m}(\bar{X} - \bar{Y})^2.$$

Under the alternative the two fitted means are the sample means. Maximizing variance gives

$$\Lambda = \left(\frac{R_x + R_y}{R_0}\right)^{(n+m)/2} = \left(1 + \frac{T^2}{n+m-2}\right)^{-(n+m)/2}, \quad T = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{1/n + 1/m}}.$$

Under H_0, $T \sim t_{n+m-2}$, because the normalized mean difference is standard Gaussian and independent of the pooled chi-square residual. Reject $|T| > t_{n+m-2,1-\alpha/2}$.

For testing equal variances with means unrestricted, the unrestricted variance MLEs are $R_x/n, R_y/m$; the restricted estimate is $(R_x + R_y)/(n+m)$. Thus

$$\Lambda = \frac{(R_x/n)^{n/2}(R_y/m)^{m/2}}{[(R_x + R_y)/(n+m)]^{(n+m)/2}}.$$

It is largest at $R_x/n = R_y/m$ and decreases in either direction. The statistic $F = (R_x/(n-1))/(R_y/(m-1))$ has $F_{n-1,m-1}$ null law. The LRT rejects for $F < a$ or $F > b$, where $\Lambda(a) = \Lambda(b)$ and $P(F < a) + P(F > b) = \alpha$. The common equal-tail F test uses quantiles $F_{\alpha/2}, F_{1-\alpha/2}$ and has exact size, but is not generally the same as the equal-likelihood LRT cutoffs. Without normality the exact t/F pivots fail: with finite variances the two-sample CLT and estimated standard errors yield a large-sample normal mean test; variance tests require finite fourth moments and a fourth-moment-adjusted standard error. Gaussian likelihood procedures under misspecification require sandwich covariance, not an automatic Wilks assertion.

33.5 Question 5

Enunciato. Compare regular vs non-regular parametric models in inference: explain what fails in the non-regular case (e.g. parameter-dependent support or boundary parameters), give at least one example from the course (e.g. Uniform$(a-b, a+b)$), and discuss consequences for MLE and LRT asymptotics (including why Wilks may fail).

Soluzione. Regularity requires an interior identifiable parameter, sufficiently smooth density on common support, nonzero finite information, and differentiation under the integral. In a non-regular model these conditions fail: moving supports introduce boundary terms, boundary parameters constrain Gaussian approximations, and nondifferentiability may prevent score expansions. A regular $\sqrt{n}$ rate and Wilks' chi-square conclusion must not be assumed.

For iid $U(0, \theta)$,

$$L_n(\theta) = \theta^{-n}\mathbf{1}_{\{M \le \theta\}}, \quad M = \max X_i, \quad \hat{\theta} = M, \quad P_\theta(M \le x) = (x/\theta)^n \quad (0 \le x \le \theta).$$

Endpoint inclusion gives the usual MLE; with a literal open support the supremum is approached from above. For $t \ge 0$ and $n > t$,

$$P_\theta(n(1 - M/\theta) > t) = (1 - t/n)^n \to e^{-t}.$$

Thus $n(\theta - M)/\theta \Rightarrow \text{Exp}(1)$, a one-sided non-Gaussian limit. Under a point null θ_0, $M \le \theta_0$ almost surely and $\Lambda = (M/\theta_0)^n$. Its exact null distribution is uniform on $(0,1)$ since $P((M/\theta_0)^n \le u) = u$.

Consequently $-2\log\Lambda$ has the χ^2_2 distribution, not Wilks' predicted χ^2_1. For $M > \theta_0$ the null likelihood vanishes and rejection is automatic. The exact level-α region is $M > \theta_0$ or $M \le \theta_0\alpha^{1/n}$.

For the shifted exponential $f_\theta(x) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$, $M = \min X_i$ and the likelihood is $e^{n\theta-\sum X_i}\mathbf{1}_{\{\theta<M\}}$. Its supremum occurs as $\theta \uparrow M$; the conventional estimator is M, and $n(M-\theta) \sim \text{Exp}(1)$. Under a point null, $-2\log\Lambda = 2n(M-\theta_0)$ has exactly χ^2_2. These examples show why calibrating automatically by χ^2_1 is incorrect. For $U(a-b, a+b)$ with known b, admissible centers lie between $M-b$ and $m+b$; the likelihood is constant there, so the MLE need not be unique. If both endpoints are unknown, the MLE endpoints are the extreme order statistics, whose errors have order $1/n$.

33.6 Question 6

Enunciato. Using the Delta method, derive the asymptotic distribution of $g(\widehat{\theta}_n)$ for a smooth map g: state the theorem, show the main steps, and explain how to build asymptotic confidence intervals and tests for transformed parameters.

Soluzione. Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, v(\theta))$ and g is differentiable at θ. Consistency and Taylor expansion give

$$g(\hat{\theta}_n) - g(\theta) = g'(\theta)(\hat{\theta}_n - \theta) + o_p(|\hat{\theta}_n - \theta|),$$

so $\sqrt{n}[g(\hat{\theta}_n) - g(\theta)] \Rightarrow N(0, g'(\theta)^2 v(\theta))$. If $g' = 0$, this first-order limit is degenerate and a higher-order expansion is needed. For a regular MLE $v = 1/I$. A plug-in standard error is

$$s_n = \frac{|g'(\hat{\theta}_n)|}{\sqrt{nI(\hat{\theta}_n)}}.$$

An asymptotic confidence interval is $g(\hat{\theta}_n) \pm z_{1-\alpha/2}s_n$. The Wald test of $g(\theta) = g(\theta_0)$ rejects when $|g(\hat{\theta}_n) - g(\theta_0)|/s_n > z_{1-\alpha/2}$, provided the derivative and information are nonzero at the relevant parameter.

For exponential rate θ, $I = \theta^{-2}$, $\hat{\theta} = 1/\bar{X}$ and $g(\theta) = \log\theta$. Since $g' = 1/\theta$, the asymptotic variance is one: $\sqrt{n}(\log\hat{\theta} - \log\theta) \Rightarrow N(0,1)$. The interval for log rate is $\log\hat{\theta} \pm z/\sqrt{n}$, and exponentiating yields the positive interval $[\hat{\theta}e^{-z/\sqrt{n}}, \hat{\theta}e^{z/\sqrt{n}}]$. MLE invariance follows by defining the profile likelihood of $\eta = g(\theta)$ as $\sup_{g(\theta)=\eta} L(\theta)$: the image of an MLE maximizes it, even for noninjective maps. Invariance concerns maximization, not equality of variances under reparametrization.